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Interrogating the research and development pipeline of artificial intelligence (AI) in health: diagnosis and prediction-based diagnosis

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Contents

Abstract

Despite the potential significant benefits of AI in health and particular in healthcare, its uptake is slow. Main barriers include trust issues, understandability of outputs and patient safety concerns. While "trustworthy AI" has been embraced as a policy programme by various global regions, issuing high-level guidance documents, it is currently unclear to which extent the clinical research community is considering these during early steps of the life cycle. To tackle this evidence gap, we systematically interrogated the biomedical literature over 6 months, focusing on the R&D phase of the life cycle in five health application areas of AI. Our analysis builds on three pillars: 1) concerned key attributes, e.g. AI solution availability, type of AI technology, clarity on interpretability considerations, medical field(s), geographic origin. 2) examined awareness and reporting of sensitive concepts related to trustworthy AI. 3) explored case studies, covering various medical fields and development stages. The current report focuses on "diagnosis and prediction-based diagnosis", summarising results of pillars 1 and 3. We found that the majority of the research articles focus on developing or fine-tuning existing algorithms, with deep learning techniques being the most commonly used. Typically, the publications did not report why specific methods had been used or whether and how interpretability was considered. Our data show considerable innovation, in particular in China, the EU and USA but highlight that clinical researchers seem insufficiently aware of critical issues including the consequences of AI techniques on interpretability. This might delay or complicate later uptake of such AI solutions.

Key message

Developing trustworthy AI in healthcare requires careful considerations of several critical key elements that impact trust during the R&D stage. The analysis of biomedical literature on AI in the area of diagnosis highlights benefits but also gaps in reporting these elements.

Executive summary

Background

While there is a considerable amount of literature on artificial intelligence (AI) in healthcare and medicine, most of these publications are either view point-type or opinion articles that present specific individual case-studies or reviews that address, on a general level, opportunities and challenges of rolling out AI in healthcare. There is however a notable gap concerning systematic evidence of the dynamics of development of AI in the scientific and clinical research communities, i.e. "research and development" (R&D) pipeline. Thus, there is little systematic information about AI developments per medical fields, to which extent AI models are developed *de novo* for clinical applications and various other aspects that would inform about the dynamics of the R&D pipeline covering the pre-deployment space of the life cycle, i.e. from conception over development, verification and validation to model assessment or evaluation within limited patient groups.

Such evidence would help understanding to which extent scientific and clinical research communities address requirements and good transparency practices that are in line with the ethical and robust translation of AI into practice, conducive to their uptake in health systems and health organisations. Key concerns include the transparency of reporting AI techniques, the extent to which explainable AI is used, in particular for visual AI which relies heavily on neural networks that are inherently 'inscrutable' without additional 'XAI' tools (XAI=explainable AI). These considerations have important ramifications for the willingness of end-user clinicians and stakeholders (namely patients) to take up AI in healthcare and accept AI's contribution to diagnostics, clinical decision-making or innovative procedures. A recent report commissioned by DG SANTE (European Commission (2022)) highlighted the slow uptake of AI in healthcare, while a review by Muehlematter UJ et al (2021) found that the EU appears to be on an equal footing with the US in regard to the number of AI systems for medical purposes (i.e. AI -based medical devices) that become available on the market.

Method of literature search and analysis

A research project was undertaken to bridge the knowledge gap concerning the dynamics of research and development (i.e. early stages of the life cycle) in various applications areas in the health sector. We used an adapted approach of the one proposed by the World Health Organization's (WHO, 2021) for categorizing these application areas. Separating diagnosis from clinical care, we use in total five (instead of WHO's four) application areas:

1. Healthcare: diagnosis and prediction-based diagnosis
2. Healthcare: clinical care
3. Health research and drug development
4. Health systems management and planning
5. Public health and public health surveillance

We selected and made use of the PubMed / MEDLINE biomedical literature database in order to collect relevant data from the publicly accessible R&D pipeline. We developed five customized search strings aligned with the five application areas outlined above, to identify AI-related research and development articles published between January 1, 2023, and June 30, 2023.

This report addresses the results only of application area one, i.e. diagnosis and prediction-based diagnosis. A forthcoming publication will describe all five application areas and include the analysis of 'key reporting parameters' captured in 18 so-called 'interrogation questions'. These were designed to explore to which extent the authors or research publications provide explicit information on key issues that are relevant for trustworthy AI, e.g. bias, data origin, intended use, validation processes etc. For detailed information see **Annex 3**.

For application area one (diagnosis and prediction-based diagnosis) we retrieved initially 230 research articles from PubMed / MEDLINE for the period chosen (6 months). Our literature screening and filtering approach (e.g. duplication removal, exclusion criteria) reduced the total number of eligible studies to 88 original research articles.

We used two approaches to analyse these data: Pillar 1 concerned six key attributes (see below), while pillar 3 concerned selected case studies, covering the three stages of the R&D pipeline outlined above as well as different medical fields.

The key attributes (pillar 1) were:

- **stage of the publication within the R&D pipeline**, ranging from development over assessment to use in research settings (including for evaluation of models).
- **availability of AI solution** (e.g. ready-to-use or customized developments)
- **AI technique**
- **Interpretability / explainability** of the AI model / system
- **medical field(s)** of application
- **geographic origin** of the publication

These attributes were chosen to explore the global innovation landscape, dynamics of specific clinical areas and selected against the background of the fact that the concept of trustworthy AI is built around ethical principles, e.g. transparency and associated concepts such as interpretability / explainability that are a key concern in healthcare, where AI systems may augment or even replace consequential decision-making by healthcare professionals.

Nr.	Attribute	Relevance for trustworthy AI, including responsible implementation and safe innovation
1	Stage within the R&D pipeline	Innovation dynamics; technical maturity; degree of clinical research versus basic model development
2	Availability of AI solution	Innovation: customized solutions versus ready-to-use AI systems. Use of large pre-trained models and transfer learning
3	AI technique	Consequences on intrinsic interpretability or need for explainability techniques. In a wider context: understandability of AI systems and their outputs in the context of informed consent by patients
4	Interpretability / explainability	• Understandability of AI systems and their outputs in the context of informed consent by patients • Potential pitfalls such as automation bias or automation complacency • Deskilling of healthcare professionals
5	Medical field(s)	AI research and development per medical area
6	Geographic origin	Global innovativeness and competitiveness in the area of diagnostics in healthcare

Findings and key conclusion

In regard to the positioning of the articles within the R&D pipeline we found that

- 80.7% of the research articles examined focused on the **development** of novel AI solutions (either from scratch or by refinement of existing algorithms)
- a smaller percentage concerned both **development and use** of AI models / systems (3.4%)
- 4.5% concerned the use of existing / already developed AI model / systems
- 9.1% focus on the use and evaluation of an AI model / system, while few articles (2.3%) are about the development, use and evaluation of an AI model / system.

The majority of AI solutions being developed for diagnosis and prediction-based diagnosis were custom AI (88.6%) (75.0%). Many of these concerned the refinement of pre-trained models (e.g. ResNet, DenseNet or EfficientNet) by transfer learning. Ready-to-use products represent 6.8% of the AI solution availability while 2.3% correspond to or a combination of custom AI and ready-to-use AI products. In some cases (2.3%), the AI solution availability is not specified (unknown).

We found that the majority of AI techniques being developed or used for diagnosis and prediction-based diagnosis correspond to deep learning (ML-DL) with an important percentage of 75.0%. However, we found that 28.8% research articles did not provide clear information on the DL technique used. Machine learning (ML w/o DL) followed deep learning (ML-DL) with 22.7%. A small percentage (2.3%) was found to correspond to hybrid techniques (ML w/o DL-DL).

Regarding machine learning (ML w/o DL), 70.0% of the articles refer to interpretable AI models while 30.0% do not have the same level of interpretability. Furthermore, we observed that there was a lack of attention to methods of explainable AI (XAI) in the case of using deep learning (ML-DL) approaches, with only 18.2% of eligible studies referring to the explainability of the AI model / system and describing additional explainability tools.

In regard to the medical field(s) of application of AI models / systems, we found that the most common fields were oncology (18.2), followed by neurology (15.9%), cardiology (13.6%), ophthalmology (12.5%), radiology (11.4%), and dentistry 6.8%), showing the dynamic of development and use of AI over various applications.

The research articles originated from a diverse set of countries and global areas, with China (42.0%), the EU (12.5%), and the USA (10.5%) contributing the largest share of publications, indicating potential research priorities and capabilities in those regions.

Overall, the research articles covered a wide range of AI model / system development, use, and assessment across various medical fields and geographical locations. We found that in some cases there was insufficient information on the precise AI technique used, a lack of transparency concerning the data used for training (in particular also in cases of 'transfer learning') and lack of detail concerning the description of AI techniques (e.g. origin, training of the AI model, etc.). The majority of the studies focused on development including further fine-tuning of existing algorithms, with a significant emphasis on deep learning techniques for diagnosis and prediction-based diagnosis. However, there was a lack of attention to explainable methods in case of development or use of deep learning approaches, with a majority of studies not mentioning explainability of techniques used. In summary many studies

- provided insufficient information about the AI techniques used,
- showed a lack of transparency regarding data origin for the training of the AI model / system,
- appeared to opt for complex (pre-trained) models without accompanying these choices with sufficient details concerning interpretability / explainability aspects.

These findings indicate a lack of awareness or sensibility of researchers to the requirements of trustworthy AI. This may delay or complicate uptake of AI solutions. Our study suggests that more effort is required to ensure that scientific and clinical research communities, early on during the development of AI systems, consider evidence needs for downstream stages of the life cycle that will facilitate trust, uptake and adoption of AI solutions in high-risk applications such as diagnostics or clinical decision making. Our findings might also indicate that the existing high-level guidance documents on trustworthy AI principles are not sufficiently detailed so that scientific and clinical research communities could translate ethical or value-based principles into practical concepts.

1. Introduction

There are high expectations on what AI could do in healthcare and medicine. However, the rapid advancement of AI applications in this sector is associated with several underexplored (and underreported) key issues, including the need to comprehend the challenges surrounding the safe and transparent implementation of AI in healthcare (OECD (2024a)). Challenges include privacy of medical and health data, sufficient robustness, discriminatory biases that may stem from a variety of sources (e.g. data, (pre-trained) models, modelling decisions), dignity of patients and healthcare professionals, interoperability problems and integration into existing workflows using legacy devices. In addition, AI requires a highly complex value chain, leading to distributed responsibilities and concerns regarding liability.

There is a plethora of perspective and opinion articles on AI in health. In addition more systematic reviews have tried to delineate in a more evidence-based manner areas of concern versus opportunities. For instance, the reviews by Subbiah V (2023), Wubineh BZ et al (2024) or Srivastava R (2024) address various ways in which AI might augment healthcare and medicine, e.g. by improving patient care and outcomes and highlight challenges concerning its safe and responsible implementation in healthcare (see also OECD (2024b)). Subbiah V (2023) highlights the need for healthcare systems to review strategies to address these challenges and fully use the potential of AI. The authors also emphasizes the importance of collaboration between healthcare professionals, policymakers, and researchers to harness the opportunities of AI in healthcare. Rajpurkar R et al (2022) discusses the potential of AI to transform the healthcare industry. While AI has shown promise in improving healthcare outcomes, enabling novel models of patient care, and reducing costs, there are also issues that need to be addressed. AI can assist with clinical decision-making, optimize healthcare workflows, and improve patient engagement. Kumar Y et al (2023) discusses the application of AI in healthcare, specifically in disease diagnosis. The paper highlights the need for large amounts of high-quality data, the risk of bias and errors, and the need for transparency and interpretability in AI decision-making. Taken together, these challenges and limitations need to be addressed for leveraging AI's full potential in health.

The general policy programme of "trustworthy AI" has gained considerable global momentum during the last years as a concept of addressing key challenges of AI in general (i.e. not sector-specific) through the formulation of ethical principles or AI principles. Particularly relevant in the present context are the documents by the EU Commission's high-level expert group, the OECD and WHO. The High-Level Expert Group (HLEG) on Artificial Intelligence set up by the Commission in 2019 published "Ethics Guidelines for Trustworthy AI" (HLEG (2019)). The Guidelines propose seven key requirements, rooted in four ethical principles, which should be addressed when developing trustworthy AI systems. These requirements are intended to inform different stakeholders partaking in AI systems' life cycle: developers, deployers and end-users, as well as the broader society. An assessment list accompanying this document was published in 2020 (HLEG (2020)). The OECD has outlined in its 2019 five "value-based" principles (OECD (2024)). Similar to the HLEG key requirements, these principles aim to outline necessary considerations for trustworthy AI. In 2023 WHO released guidance on ethics and governance of AI for health (WHO (2023)), a comprehensive document built around WHO's own six ethical principles that relate to the bioethical principles of Beauchamp TL and Childress JF (1979); beneficence, non-maleficence, justice and fairness, autonomy. Finally, the EU is the first global region to implement a comprehensive legal framework for AI, the "AI Act", which entered into force in 2024 (AI Act (2024)).

Despite the potential benefits of AI in healthcare and the policy programme of “trustworthy AI”, the uptake of AI technologies in the health sector is slow (European Commission (2022)) and inconsistent in regard to health settings and / or health systems, leading to health inequality (Mittelstadt B (2021)).

There is ongoing debate about the reasons, with efforts trying to identify specific barriers. A recent systematic review of Ahmed MI et al (2023) identified seven barriers: ethical, techological, liability, workforce social and patient safety, with some of these overlapping. For instance, an ethical barrier is the lack of transparency and specifically interpretability of algorithms. This issue is also referred to as the ‘black box’ or *opacity* problem Chen J et al (2018). Lack of transparency and specifically interpretability (i.e. why and how an algorithm generated a specific output / recommendation) may have negative consequences on validation, clinical oversight, patient-physician relationship, but which may also lead to safety issues. Lack of transparency is one of the major factors that undermines the trust necessary for adoption; other factors are privacy and accountability. Ensuring demonstrable patient safety and sufficient understandability of AI outputs is paramount for the successful and responsible implementation of AI in healthcare. The barriers to AI implementation discussed in this review are applicable across various medical fields. Overcoming these would likely accelerate responsible use of AI in health (OECD, 2024), providing potential significant benefits to healthcare.

These barriers need to be considered and addressed not only at late stages of the AI life cycle, but, ideally from the very first stages, i.e. conception, development and verification / validation. We refer to these three stages as the ‘development & research’ pipeline. At this stage critical decisions will be taken, e.g. whether to select a complex over a simple model, whether to use a pre-trained large model, whether to use an intrinsically interpretable model (e.g. random forest) over an intrinsically ‘inscrutable’ model (e.g. convoluted neural network). Other key decisions relate to data collection and compilation and associated privacy issues. We refer to this process towards a usable AI system as ‘algorithm-to-model transition’ (Griesinger et al, in preparation). Taking decisions during this process in circumspect way with regard to ethical, technological and patient safety issues is critical for building a ‘trustworthy’ algorithm or AI system.

In order to investigate to which extent researchers at this upstream stages are considering above elements, we designed a systematic study of the ‘R&D pipeline’ in health, using ‘key attributes’ selected against the background that trustworthy AI is built around ethical principles. Another motivation for the study was to determine the global innovation landscape and the dynamics per medical fields. For categorizing areas of application, we adapted the WHO’s approach, leading to five areas. This report focuses on the most dynamic area, i.e. diagnosis and prediction-based diagnosis. Utilizing the general PRISMA workflow, we carefully designed five search strings for retrieving relevant publications and devised relevant inclusion / exclusion criteria (‘filtering approach’). Scrutinizing the PubMed / MEDLINE database over a period of six months, we retrieved 88 original research publications in the area of diagnosis.

This is, to our knowledge, the first study that systematically examines the research and development pipeline for AI health applications. It is particular timely given that the OECD has published, this year, two important briefings on AI in health (OECD (2024); OECD (2024b)), emphasizing the need to put ethical principles into practice in order to leverage the significant potential of AI in healthcare while adequately identifying and controlling risks and calling for stronger efforts for collaboration. This study benefitted from the JRC’s expertise in the area of medical devices and innovative devices requiring particular scrutiny.

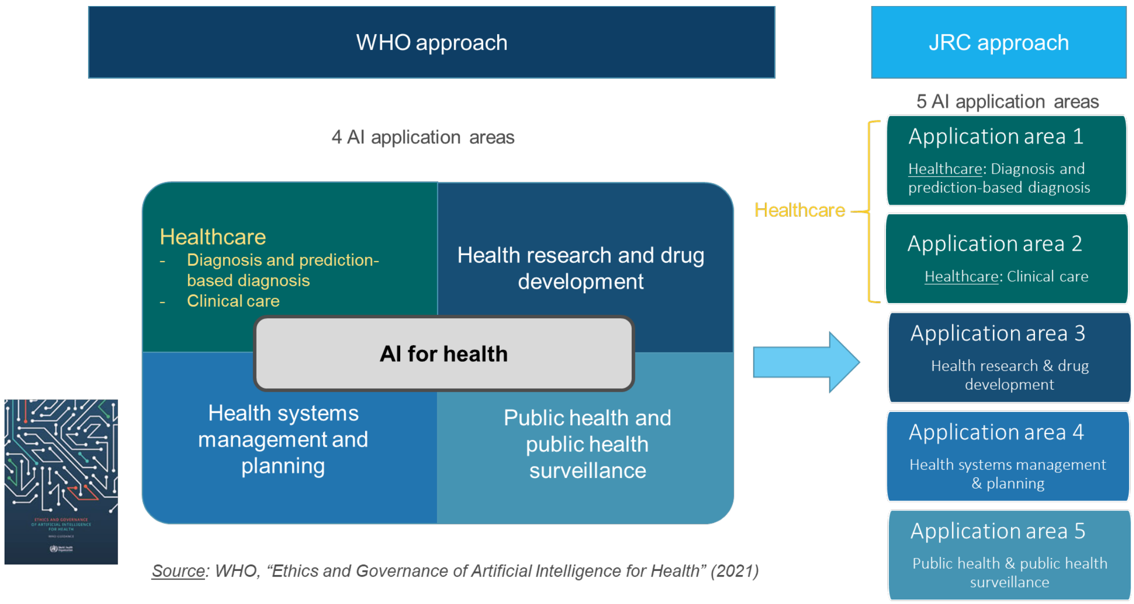
2. Methods

We explored the available scientific literature with regard to our research topic by applying a certain literature search strategy as explained in detail below. To facilitate the retrieval and categorisation of the relevant information, the literature search strategy was applied separately to pre-selected AI health application areas.- We used an adapted approach of the one proposed by the World Health Organization's (WHO, 2021) for categorising these areas. We use in total five (instead of WHO's four) application areas:

1. Healthcare: Diagnosis and prediction-based diagnosis
2. Healthcare: Clinical care
3. Health research and drug development
4. Health systems management and planning
5. Public health and public health surveillance

For illustrative purposes these selected application areas in the field of health are depicted in Figure 1 in comparison to the 2021 WHO approach.

Figure 1. WHO / JRC classification approach of the AI application areas in the field of health



Source: JRC/Own production

Furthermore, we developed a stepwise systematic approach for the analysis of the eligible research articles by interrogating and exploring their content based on the following three pillars:

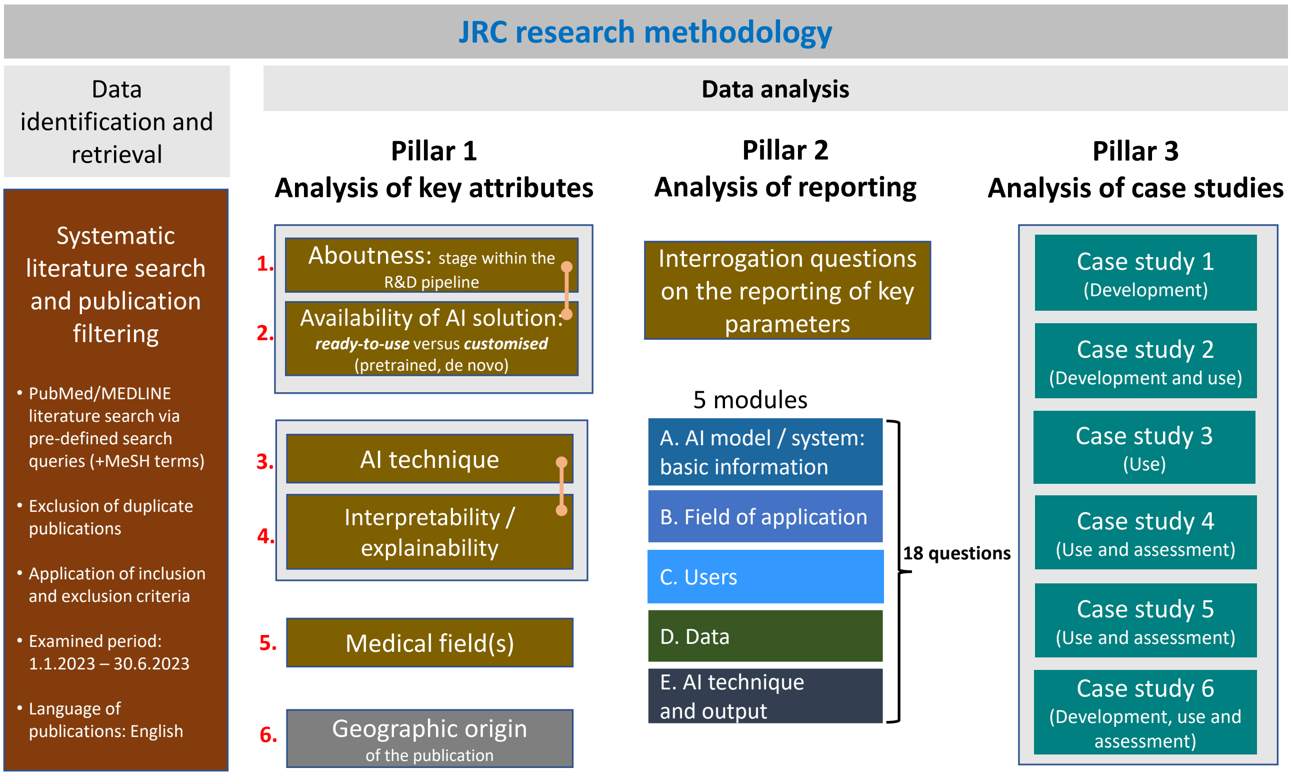
Pillar 1: In this first pillar we aimed to identify, collect, and analyse a set of key attributes (1-5). These attributes were chosen to explore the global innovation landscape, dynamics of specific clinical areas and selected against the background of the fact that the concept of trustworthy AI is built around ethical principles. These key attributes were: stage of the publication within the "research and development" (R&D) pipeline ("aboutness"), availability of AI solution, described AI technique, interpretability / explainabilty of the AI model / system, medical field of application and geographic origin of the study.

Pillar 2: In this second pillar we proceeded with an analysis of the reporting of certain key parameters with the aid of a set of 18 interrogation questions that we developed specifically to this purpose. These questions were organised in five AI-related thematic modules (A-E). This pillar examines the sensitivity of the scientific and clinical research communities in regard to providing explicit information on key issues that are relevant for trustworthy AI. These 5 modules are: basic information on the AI model / system, field of application, users, data, and AI technique and output.

Pillar 3: Finally, we selected and presented tangible examples that cover the three stages of the R&D pipeline ("aboutness"), also in regard to medical fields.

The above described research methodology approach is schematically presented in Figure 2 while more detailed information is provided below in this section of this report.

Figure 2. The three-pillar research methodology approach



Source: JRC/Own production

2.1 Systematic literature search and publication filtering

2.1.1 Sources of scientific information

PubMed / MEDLINE was identified as the most appropriate bibliographic information source, i.e. scientific literature database with regard to the subject area and the aim of this research project (for more details see **Annex 1**).

2.1.2 Scientific literature search protocol

Our scientific literature search and data retrieval strategy initially comprised of the development of a scientific literature search protocol based on certain PubMed / MEDLINE search strings / queries that were carefully formulated according to the aim of our study (see below and **Annex 2**). The first results from the literature search included various types of publications, e.g. original research articles, reviews (including systematic reviews), editorials, perspectives, view articles, letters to the editor, opinions, comments, etc. At the next stage these results were further filtered by following a specific scientific literature filtering approach so that in the end only original scientific research articles relevant to the 5 application areas were eligible for the results analysis stage (see relevant section 2.1.3).

Development of PubMed / MEDLINE search strings

We developed a unique PubMed / MEDLINE search string for each of the 5 application areas in order to acquire the most relevant publications and data for each application area.

The literature search terms used in each search string were carefully selected and included both Medical Subject Headings (MeSH) Terms and appropriate keywords All search terms were combined to form the complete search strings using the Boolean terms “AND”, “OR”, “NOT”.

The detailed PubMed / MEDLINE literature search queries developed for application area 1 is shown in **Annex 2**. The same approach was used for other application areas (2-5), however these areas are not subject of this report.

Conduct of systematic literature search

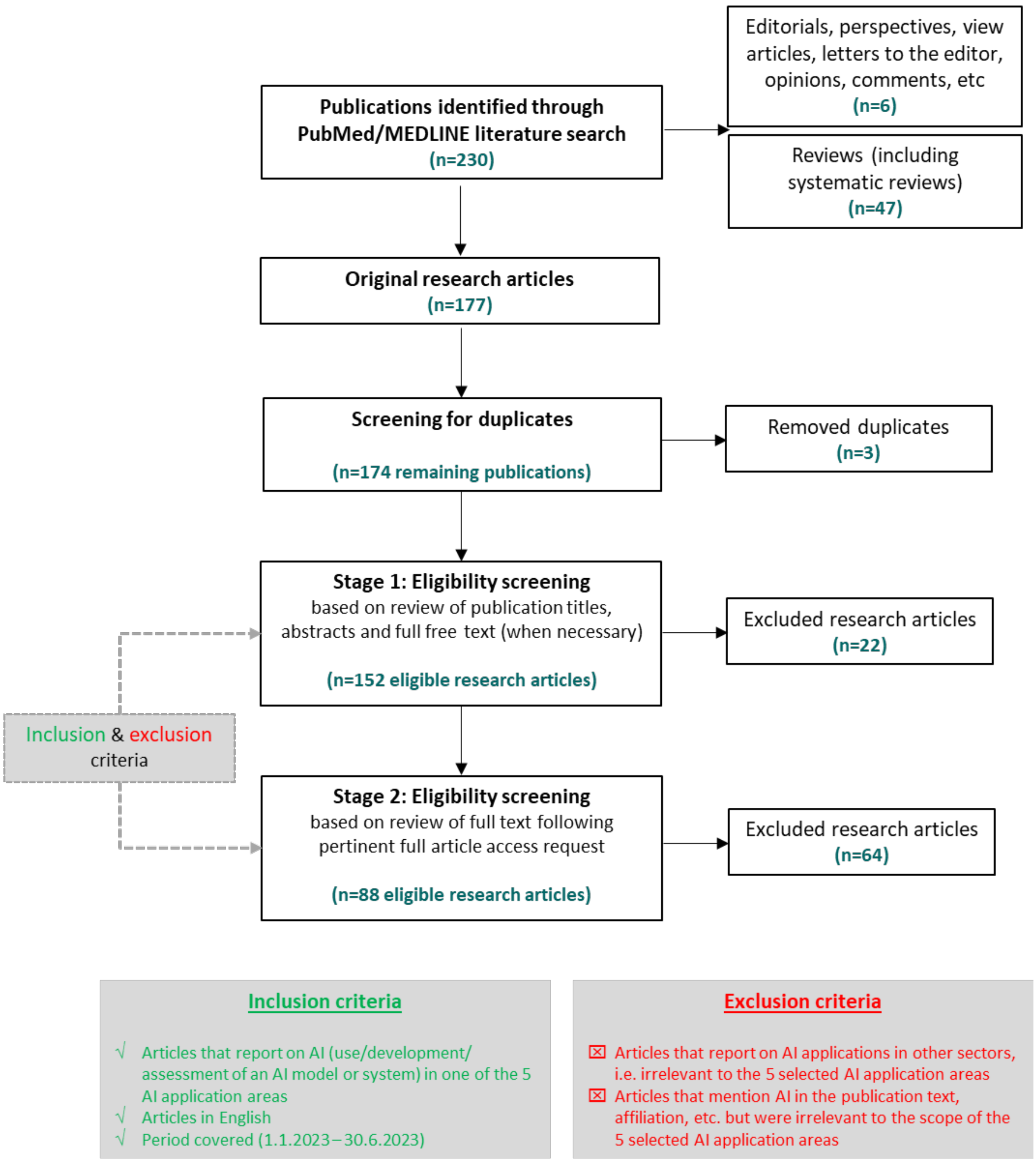
A systematic scientific literature search available in PubMed / MEDLINE was conducted for AI-related scientific research articles published during the period January 1, 2023, and June 30, 2023 in the English language. The literature search was performed on July 13, 2023 for application area 1 and on July 18, 2023 for application areas 2, 3, 4 and 5. According to the PubMed User Guide, the PubMed date filter used to select our results (custom time range) counted all publication dates for a citation as supplied by the publisher. This included both electronic (online versions and advanced / ahead online versions of articles available before they are filtered into a particular journal issue) and print publication dates.

The results from each literature search were exported from PubMed / MEDLINE and saved in tabular format (*.csv) in an Excel spreadsheet. Furthermore, the publications identified in the systematic search were uploaded to EndNote reference manager and duplicates were automatically identified and deleted.

2.1.3 Scientific literature filtering approach

A comprehensive selection of the eligible research articles was carried out in two distinct stages by implementing a literature filtering approach (Figure 3) as explained in more detail below.

Figure 3. Scientific literature search protocol and filtering approach for application area 1



Source: JRC/Own production

Stage 1: Filtering of the retrieved publications by abstract and title

Once all literature search results were obtained from the initial literature search in PubMed / MEDLINE, a staged literature filtering approach was applied in order to exclude irrelevant publications and identify only the eligible articles for this research project, i.e. only original AI research articles in the relevant AI area under examination. The publications filtering was an essential stage as initially no filters had been applied in the advanced PubMed / MEDLINE search, which led to the retrieval of other various, non-eligible, types of articles, e.g. reviews (including systematic reviews), editorials, perspectives, view articles, letters to the editor, opinions, comments, etc.

To this end, we dedicated time and effort to filter manually either by article type (as labeled) examination or by content review all the various publications retrieved initially from the PubMed / MEDLINE search. This is because PubMed filters apparently were not always very efficient to distinguish between original research articles and other publication types which were not properly labelled.

In this stage a careful screening of the initially retrieved publications by title and abstract was performed. In case the content of title and abstract was not considered conclusive to decide on the eligibility of the article, a further screening of the full text of the article was performed when it was publicly available (free access publications). Where there was no free access to the full text of the publication we requested full access to it via the JRC library in order to examine the full paper at the next stage (see stage 2).

Stage 2: Further filtering of the retrieved literature with focus on certain AI-related inclusion criteria

During stage 2 we screened the full text of all publications to which access to the full text of the publication needed to be requested from the JRC library service so that we could decide on the eligibility of the article when this not was possible from the title / abstract review in stage 1.

Furthermore, in this stage we carefully reviewed the full text of all research articles selected so far by additionally applying certain AI-related inclusion criteria in order to decide on the final eligibility of each research article. This was a crucial part of the filtering approach which defined the total number of eligible studies for each application area. The research articles were assessed based on the content and the contribution of the full publication text to the development of the corresponding AI-related research area of interest for this project.

<u>Inclusion criteria</u>

In particular, the following **inclusion criteria** were applied for considering as eligible each research article per application area:

- Research articles dealing with AI applications in the diagnosis and prediction-based diagnosis (→ eligibility criterion for application area 1)
- Research articles dealing with AI applications in clinical care (→ eligibility criterion for application area 2)
- Research articles dealing with AI applications in health research and drug development (→eligibility criterion for application area 3)
- Research articles dealing with AI applications in health systems management and planning (→eligibility criterion for application area 4)
- Research articles dealing with AI applications in public health and public health surveillance (→eligibility criterion for application area 5)

Exclusion criteria

Articles that did not include any of the above or had insufficient information regarding AI applications were excluded. Furthermore, publications where AI was briefly mentioned but did not report on AI (e.g. in the text, affiliation) or mentioned AI but for irrelevant to this project purposes were also excluded. Finally, articles with abstract or full text not written in the English language were excluded as well.

In all stages we applied the "**4 eyes Principle**" regarding the critical assessment (screening and filtering) of the scientific literature. To this end two reviewers assessed independently all scientific publications retrieved from PubMed / MEDLINE according to the abovementioned methodology. In case of discrepancies during their assessment the two reviewers put effort to resolve it through discussion and reach consensus. In case of persisting disagreement the reviewers could consult a third reviewer (project leader) in order to make the final decision.

Following the above described literature search protocol we managed to retrieve initially 230 articles from PubMed / MEDLINE for application area 1. A total of 177 studies were screened after removal of duplicates. Eligibility screening based on title and abstract led to a total of 152 studies. Each full-text articles was screened reducing the total number of eligible studies to 88 (eligible research articles). The application of the literature search protocol along with the implemented literature filtering approach is schematically shown in Figure 3.

2.2 Systematic analysis of the eligible research articles

We analysed the eligible original research articles using a systematic approach, i.e. in three pillars.

2.2.1 Pillar 1: Analysis of key attributes

In the first pillar we analysed the eligible research articles with regard to certain fundamental publication key attributes related to the global innovation landscape, i.e. positioning of the study along the R&D pipeline ("aboutness"), availability of the AI solution, described AI technique, interpretability / explainabilty of the AI model / system, medical field of application and geographic origin of the study (based on certain criteria regarding the geographic origin of the authors and / or their affiliation).

Below more detailed information is provided on each of the publication attributes we selected to analyse for the purpose of this project.

1) Aboutness: stage within the R&D pipeline

The focus of the research publications is mostly on the early stages of the life cycle of AI model / systems, which refer to as the R&D pipeline. This pipeline includes stages such as concept, development, verification and validation, but does not extend to deployment and post-deployment.

With 'aboutness' we describe the positioning of the AI technology described in the research article along this simplified pipeline covering the following "aboutness descriptors":

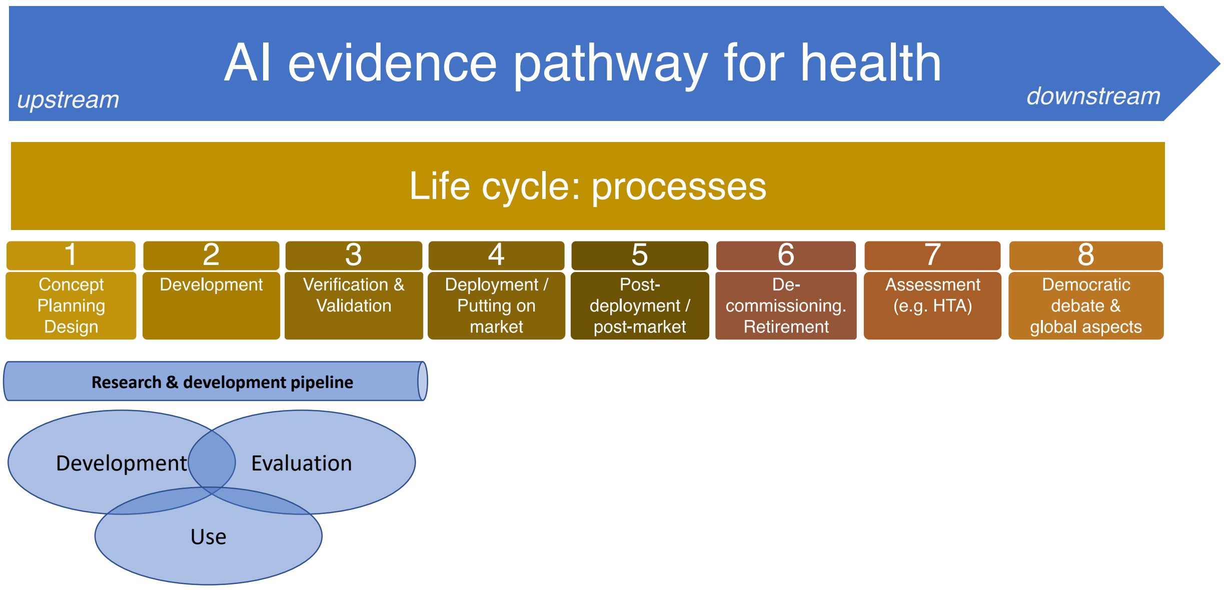
i) **Development** of a model or AI system,

ii) **Use** of an already available and developed model or AI system for specific purposes (e.g. diagnostic problem / question), and

iii) **Evaluation (including post processing)** of various aspects of an available model or AI system, e.g. comparisons between AI system and human readers or comparisons between AI systems, including evaluation for independent patient groups. Volkinger K et al (2021) mentions that the evaluation step involves assessing the performance of the model on independent groups of patients. This means that the model is evaluated on patient data that was not used to train the model, and that is representative of the population that the model will be applied to in clinical practice.

Figure 4 illustrates the R&D pipeline, including the stages of concept, design, development, verification, validation, and deployment of an AI model or system.

Figure 4. Aboutness: stage within the R&D pipeline



Source: JRC/Own production

We used the above three **aboutness descriptors** to label the eligible research articles based on the concept explained in detail in Table 1. Importantly, we make the assumption that a given research article may be described / presented by more than one aboutness descriptors.

Table 1. Aboutness descriptors and their concept

Nr.	Aboutness descriptors	Concept
i	**Development**	• This descriptor is used for publications that report on the **development** of an AI model or AI system or several of these. • This involves the use of any machine learning algorithm (either developed from scratch or retrieved from other parties) that is run on data ("training") to yield a model for a specific real-world problem. • The descriptor also covers situations where pre-trained models were subjected to **"transfer learning".** • The descriptor also includes **model validation and model testing**, commonly seen as part of model development. Notably, model validation includes the fine-tuning of parameters (e.g. through "calibration").

ii	**Use**	• This descriptor is used for publications that **make use** of an AI model or AI system or several of these and where the publications describe the application to a relevant real-world problem (e.g. area of diagnosis) **beyond model validation and testing** (see descriptor "development"). • This includes publications that make use of commercial or otherwise available AI models or AI systems to address a specific use case.
iii	**Evaluation (including post-processing)**	• This descriptor is used for publications that not only use an AI model or AI system or more than these for a given application but also entail aspects of **evaluation of the performance and / or other aspects (e.g. safety, robustness) under use conditions for a given real-world problem**. • This includes for example **comparative evaluations** of different AI models or AI systems for a given purpose or the **evaluation of an AI model or AI system in comparison to the performance of human readers**.

Source: JRC/Own production

2) AI solution availability

The availability of AI solutions in healthcare and medicine is increasing rapidly, with many AI-powered systems being developed and deployed in various clinical settings. However, there are also challenges regarding the adoption of new or novel AI solutions.

We analysed the eligible research articles with regard to the availability of AI solution that has been developed or used for the given application area. The AI solution may include custom AI (untrained or pre-trained, *de novo*) or ready-to-use AI product / commercial or a combination of the two.

3) AI technique

AI technique description is an essential aspect of ensuring that AI models / systems are reliable and trustworthy. AI encompasses a wide range of techniques aimed at enabling machines to simulate human-like intelligence and behavior. Some common AI techniques are machine learning (ML w/o DL), deep learning (ML-DL), natural language processing (NLP), etc. These are just a few examples of AI techniques, and there are many more specialised techniques, e.g. subsets of DL such as convolutional neural network (CNN) or deep neural network (DNN). AI research and development continue to evolve rapidly, with new techniques and advancements emerging regularly.

For the purpose of our study we analysed the eligible research articles with regard to the type of AI technique that has been employed in the AI model / system.

4) Interpretability / explainabilty of the AI model / system

When implementing AI applications in the healthcare domain a key concern is the extent of interpretability or explainability afforded by the models, i.e. to which extent their general functioning and, in particular, how and why they generate specific outputs. This is critical for maintaining informed consent by patients, but also to understand sources of errors or problems (e.g. misclassifications) and be able to trace them with the ultimate aim of rectifying the situation.

Many AI models, in particular neural networks / deep-learning are not readily interpretable ("black-box" models). While their training data may be transparent and while their predictions may be accurate, there is no easy measure for understanding why a model is making a specific prediction.

This can be a particular issue in the area of diagnosis, where it is essential to understand the reasoning behind a diagnosis. When the intention is to develop and use a trustworthy AI model, the implemented

AI technique cannot be left in a “black box” state, i.e. the black box needs to be ‘opened’ through various explainabiltiy techniques allowing its developers, patients, users (in particular healthcare professionals) and stakeholders to understand how the AI model or system works in order to trust its results.

In fact the interpretability / explainability is one of the most heavily debated topics when it comes to the applications of AI in healthcare (Amann J et al (2020)). Recent review articles emphasise the importance of interpretability / explainability of AI systems (Holzinger A et al (2019); Linardatos P et al (2021); Frasca M et al (2024)). These reviews aim to explore the growing impact of machine learning and deep learning algorithms in the medical field, specifically focusing on the critical issues of explainability associated with black-box algorithms. This could lead to the development of more interpretable AI models, enabling clinicians to understand the reasoning behind decisions. The reviews propose distinct concepts for interpretability and explainability, aiming to clarify their implications for the health decision-making.

- *Interpretability* refers to the human ability to understand the functioning and decision-making process of an AI model (understanding or gaining insight into how it works). Interpretability is often used in the context of models that are intrinsically understandable (or “interpretable by nature” (Burkart N and Huber MF (2021)), such as decision-trees, rule-based models or linear models. However, transparency on the algorithm or the combination of algorithms employed in the AI model / system is a prerequisite for interpretability.

- *Explainability:* The development and/or use of more complex AI models (e.g. deep learning, convolutional neural networks (CNN), graph neural networks (GNN), etc.) present some advantages. However, such models can be difficult to be interpreted (so-called ‘inscrutable’ (Weld DS and Bansal G (2018) or ‘not intrinsically interpretable models’) and require additional tools for deducing explanations about their functioning and how and why specific outputs are generated under given circumstances. In the area of diagnosis, this is particularly the case of computer vision which relies heavily on neural networks that are impossible to understand / interpret without additional explainable artificial intelligence (XAI) tools. This approach is captured by the term “explainability”.

In our study, information was identified and collected regarding the clarity on interpretability / explainabilty of the AI model / system presented in the publication:

- For models considered interpretable, our analyses focused on whether the publication provided sufficiently clear information about the algorithm(s) – a prerequisite for interpretability “by nature”.

- For ‘inscrutable’ AI techniques, we examined whether the publication provided information about relevant techniques to render the model explainable, captured under the efforts termed “explainable AI (XAI)”.

5) Medical field of application

The medical field is an essential element as it shows the dynamic of development and use of AI over various applications in healthcare and medicine. Different medical fields have different requirements, challenges, and nuances as well as different clinical contexts, including the types of patients, diseases, and treatments.

We identified and collected the available information regarding the medical field for which the AI solution has been developed, used or assessed according to the publication. To this end, we developed a customised list with pre-defined medical fields that we used in order to assess and match the most relevant medical field with the available information in the publication. Our customised list with the pre-defined medical fields was largely based on the following two pre-existing lists:

a) the list with the medical fields (medical specialties) stated in Annex V of Directive 2005/36/EC on the recognition of professional qualifications in the European Union *(https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A02005L0036-20211210)* and

b) the list with the medical specialties mentioned in the webpage of the European Union of Medical Specialists (UEMS), i.e. *www.uems.eu/our-structure#Medical-specialties*.

Background information: The aim of Directive 2005/36/EC was to coordinate the system of mutual recognition of professional qualification in EU/EEA countries. The European Union of Medical Specialists (UEMS) is a professional organisation of doctors presenting medical specialists in the EU. UEMS collaborated with the Commission on the Directive 2005/26/EC on Professional Qualifications (and its following updates). The most important medical specialties bodies within the UEMS are listed on its website. Due to a substantial overlap between some of the specialties (for example "Emergency Medicine" and "Accident and emergency medicine" refer to a large degree to the same pattern of practice across the EU) we merged the two lists and provided a final list of medical fields that we used in the present research project (please see ***Annex 4****).*

6) Geographic origin of the publication

Another publication key attribute is the geographic origin of the publication, i.e. the location of the affiliation of the authors of the published research articles. It is essential to identify and understand potential clusters in regard to AI development and use and especially with regard to potential uptake in the clinical context.

Information on the geographic origin (country) of the author(s) of all publications was identified taking into account the following three aspects related to the academic affiliation of the authors:

i) location of the academic affiliation of the corresponding author(s),
ii) location of the academic affiliation of last (senior) author,
iii) location of the academic affiliation of the majority of authors.

In cases where different locations were stated in the publication for the above aspects the final decision on the geographic origin of the publication was made based on the alignment of at least two of the above aspects.

2.2.2 Pillar 2: Analysis of reporting of key parameters

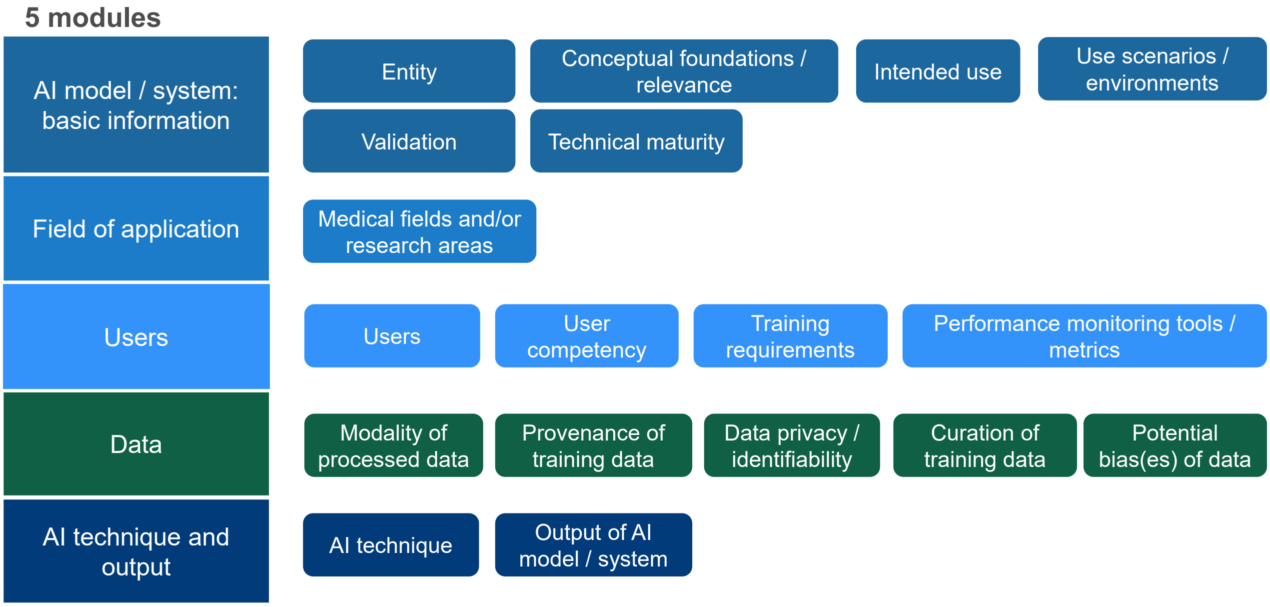
Reporting of key parameters following the development and use of interrogation questions

It was decided to of establish a list of key parameters for which their reporting in each eligible research article was considered as essential to identify and critically assess the research study.

For this purpose existing AI frameworks across sectors were analysed and compared in details as a basis for the identification of a consistent set of publication key parameters of AI system applications. For example, the OECD framework describes and classifies AI systems and applications along the following dimensions: People & Planet, Economic Context, Data & Input, AI Model and Task & Output. Each one has its own properties and parameters (OECD, 2022).

We developed a relevant comprehensive set of 18 interrogation questions with three fixed possible answers that would enable the reviewers to assess in a systematic way the level of reporting of the key parameters in each eligible research article. Key parameters along with their corresponding interrogation questions were organised in five AI-related thematic modules as schematically shown in Figure 5.

Figure 5. The AI-related thematic modules and the associated key parameters



Source: JRC/Own production

<u>Content of interrogation questions per module</u>

As already explained the 18 interrogation questions were assigned to five (5) AI-related thematic modules. These 18 interrogation questions aimed to facilitate the reviewers to determine the level of reporting on the selected key parameters in each research article. More information on the content of the interrogation questions per module can be found below.

A) Module: Basic information on the AI model / system

This module focuses on the identification and examination of essential information in the eligible research articles on the entity responsible for developing / using the AI model / system, and its conceptual foundation / scientific relevance, intended use, use scenario / environment, validation and technical maturity. In particular we examined each research article for the reporting of the following key parameters:

A.1. Entity

The research article was investigated regarding the reporting of information about the entity responsible for developing and / or providing the AI model / system.

A.2. Conceptual foundations / scientific relevance

The research article was investigated regarding the reporting of information on the scientific evidence in relation to the AI model / system's design and its intended use. For application area 1 this includes for instance the relevance of a biomarker or anatomical structure for diagnostic predictions or the established clinical value and / or effectiveness of a specific algorithmic outcome.

A.3. Intended use

The research article was investigated regarding the reporting of information on the intended use of the AI model / system.

A.4. Use scenarios / environments

The research article was investigated regarding the reporting of information on the specific use scenario(s) and use environment(s) in which the AI model / system is intended to operate.

A.5. Validation

The research article was investigated regarding the reporting of information on the extent of the initial quality evaluation of the AI model / system.

A.6. Technical maturity

The research article was investigated regarding the reporting of information on the technical maturity of the AI model / system.

B) Module: Field of application

This module focuses on the identification and examination of essential information in the eligible research articles on the medical field and / or the research area related to the described application of the AI model / system.

B1. Medical field and / or research areas

The research article was investigated regarding the reporting of information on the medical field(s) and / or research areas to which the AI model / system would apply or / contribute.

C) Module: Users, operators, monitoring tools and metrics

This module focuses on the interrogation of the eligible research articles on the reporting of the following key parameters related to the training of the users with regard to the intended use of the described the AI model / system:

C.1. Identification of potential users

The research article was investigated regarding the reporting of information on the clear identification of the potential users of the AI model / system.

C.2. User competency

The research article was investigated regarding the reporting of information on the required user competency.

C.3. Training requirements

The research article was investigated regarding the reporting of information on training requirements for the competent and safe intended use of the AI model / system.

In the context of AI models in healthcare (application area 1), training requirements typically involve:

- Understanding the capabilities and limitations of the AI model.
- Knowing how to operate the model and interpret its outputs (e.g., Class Activation Mapping).
- Integrating the AI model's results with clinical knowledge and patient data for decision-making.
- Adhering to relevant safety protocols and privacy regulations when handling patient data.

C.4. Performance monitoring tools / metrics

The research article was investigated regarding the reporting of information on the availability of performance monitoring tools / metrics that should be used by the operators for the AI model / system.

D) Module: Data

This module focuses on the interrogation of the eligible research articles on the reporting of the following key parameters related to the data acquired / used with regard to the training / use of the described AI model / system:

D.1. Modality of processed data

The research article was investigated regarding the reporting of information on the data modality or modalities processed by the AI model / system.

D.2. Data provenance of training data

The research article was investigated regarding the reporting of information on the provenance of the data employed for training the AI model / system.

D.3. Privacy / handling of personal data

The research article was investigated regarding the reporting of information on whether and how personal data issues have been addressed during the development or use of the AI model / system.

D.4. Curation of training data

The research article was investigated regarding the reporting of information on the curation of the data employed for training the AI model / system.

D.5. Potential sources of data bias

The research article was investigated regarding the reporting of information on potential sources of bias associated to the data handled by the AI model / system.

E) Module: AI technique and output

This module focuses on the interrogation of the eligible research articles on the reporting of the following key parameters related to the AI technique and the AI model output:

E.1. AI technique

The research article was investigated regarding the reporting of information (clear description) on the AI technique employed, e.g. machine learning (ML w/o DL), deep learning (ML-DL), convolutional neural network (CNN), graph neural network (GNN), etc.

E.2. Output of AI model / system

The research article was investigated regarding the reporting information on the generated output(s) of the AI model / system.

Methodology on the assessment of the interrogation questions

All interrogation questions on the reporting of information regarding the above presented key parameters were answered by selecting one of the three following pre-defined options, i.e.:

a) **"Clear information"**, when clear information was reported in the research article regarding the key-parameter under investigation

b) **"No information"**, i.e. there was no information reported in the research article regarding the key parameter under investigation

c) **"Information with uncertainty"** i.e. there was limited or not entirely clear information reported in the research article regarding the key parameter under investigation.

A summary table with all interrogation questions used in our study for the assessment of the reporting of the key parameters is shown in **Annex 3**.

Methodology on the assessment of the technical maturity of AI systems

In the case of a positive answer (reporting of clear information) provided to question A.6 (Technical maturity) the reviewers attempted to make a subjective assessment with regard to a three level classification of the technical maturity of the AI model or AI solution described in the research article.

The technical maturity of AI systems can vary widely. Technology readiness levels (TRLs) can help classify an AI system's technical maturity. OECD (2022) describes 9 TRL categories. These categories were based on the JRC analysis conducted by Martinez Plumed F et al (2020) (JRC129399) on the levels of maturity and readiness of newly introduced technologies.

For the purpose of our study, we simplified the classification and merged the different OECD categories in 3 distinct stages, as follows:

- **M1 (concept stage):** from basic principles observed to experimental proof of concept (TRL 1 to 3)
- **M2 (research / validation stage):** from technology validated in the lab to technology demonstrated in a relevant environment (TRL 4 to TRL 6)
- **M3 (deployment stage):** from system prototype demonstration in operational environment to actual system proven in operational environment (TRL 7 to 9).

In the light of the above when interrogating the research articles with question D.6 on the technical maturity of the described AI system, we provided the following three options for selection and further statistical analysis for the purpose of our project:

a) M1 (concept stage)
b) M2 (research / validation stage)
c) M3 (deployment stage)

Use of the interrogation questions according to the 4 eyes principle

We made use of the table with the interrogation questions and the relevant guidance text (see **Annex 3**) by following again the "**4 eyes principle**", i.e. two reviewers assessed (interrogated) independently each eligible research article and selected one of the available three fixed answers, i.e. clear information, no information, information with uncertainty.

In case of any discrepancies found during the assessment the two reviewers put effort to resolve it through discussion and consensus. In case of persisting disagreement (usually in complex cases) the reviewers could also request the assistance of a third reviewer (the project leader) to facilitate the final decision.

For the assessment of the eligible research articles falling under application areas 1-3 the whole set of the 18 interrogation questions was used. For application area 3 (health research and clinical development), some questions were considered, because of their nature, not applicable to the content of the publication in the given application area, i.e. question D.2 on provenance of training data, question D.3 on privacy / handling of personal data and question D5 on data-based bias in some cases e.g. AI applications for drug design and development.

Similarly for application areas 4 and 5 no assessment was performed at all based on the interrogation questions as the latter were considered as not applicable to the content of the publications in these two areas.

The extent of use of the set of the 18 interrogation questions across the 5 application areas is illustrated in a summative way in Table 2:

Table 2. Interrogation questions used per AI health application area

Application area	Interrogation questions
1. Healthcare: diagnosis and prediction-based diagnosis	All
2. Healthcare: clinical care	All
3. Health research and clinical development	All except questions D2, D3 and D5 (in certain cases
4. Health administration / health systems management & planning	N/A
5. Public health and public health surveillance	N/A

Source: JRC/Own production

2.2.3 Pillar 3. Analysis of case studies

In this stage, we explored the dynamics of AI developments in the different areas of application through case studies. We made a selection of selected tangible examples that cover the three stages of the R&D pipeline ("aboutness"), also in regard to medical fields.

Our selection of examples cover the

- Development,
- development and use,

- use,
- use and evaluation,
- development and use and evaluation

of an AI model / system.

For each case study we looked for and collected information the research articles on the following elements:

- Bibliographic information on the published research article
- Description of the rationale / summary of the study based on the available information in the abstract, introduction, methodology section of the research article
- Description of the main findings and conclusions as presented by the authors in the results and conclusion sections of the research article
- Summary of the limitations of the study stated by the authors of the research articles
- Description of the key attributes and reporting of key parameters

All the above information was systematically collected and documented in a “fiche template”, (see **Annex 5**) that we developed for this purpose.

2.3 Data analysis and visualisation

The extracted data from the analysis of the research publications were analysed and visualised with Qlik Sense® (by Qlik, www. Qlik.com), a powerful data analytics platform that is now available to different Commission DGs.

Qlik Sense can create personalised, interactive data visualisations, reports, and dashboards from multiple data sources with drag-and-drop ease.

In particular, Qlik Sense is a client managed solution (in contrast to the Qlik Cloud Analytics - a cloud based-solution) which has been designed for on-premises use in highly regulated professional environments like the Commission’s DG’s.

3. Results

We present the outcomes of our analysis of the research articles that were classified under the application area 1, i.e. diagnosis and prediction-based diagnosis. The results of are presented per pillar below and in the same order we explained the corresponding research methodology aspects in the previous section (2). This report is focusing on the results of our analysis of key attributes ((pillar 1) and on case studies (pillar 3). A forthcoming study will present the results of all three pillars for all five of the application areas (Chassaigne H et al., in preparation).

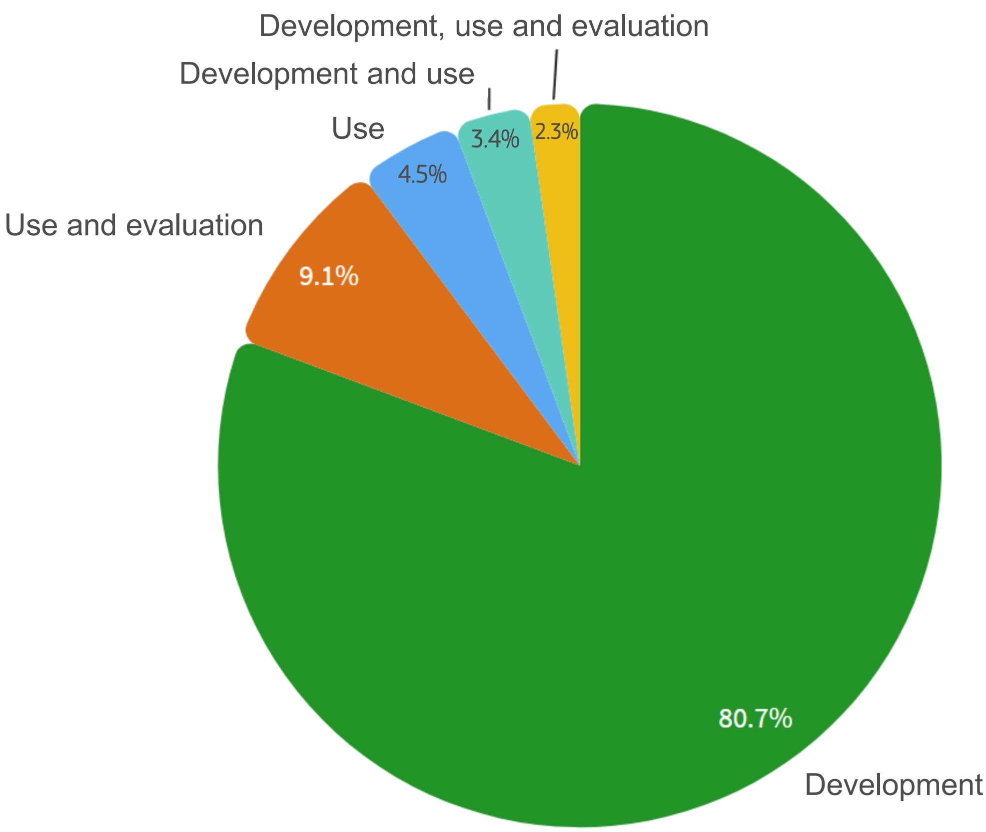
3.1 Pillar 1: Analysis of key attributes

This section aims to demonstrate the outcome of the descriptive statistical analysis of certain fundamental attributes related to the global innovation landscape, i.e. the aboutness, AI solution availability, AI technique, interpretability / explainability of the AI model / system, medical field of application, as well as the geographic origin of the publication.

3.1.1 Aboutness: stage within the R&D pipeline

We collected and identified information regarding the aboutness of the research article, i.e. the positioning of the AI technology along a simplified R&D pipeline. We used the three pre-defined aboutness descriptors to label and categorise the research articles (please see method section 2.2.1 for more details on their concept).

Figure 6. Stage within the R&D pipeline



Source: JRC/Own production

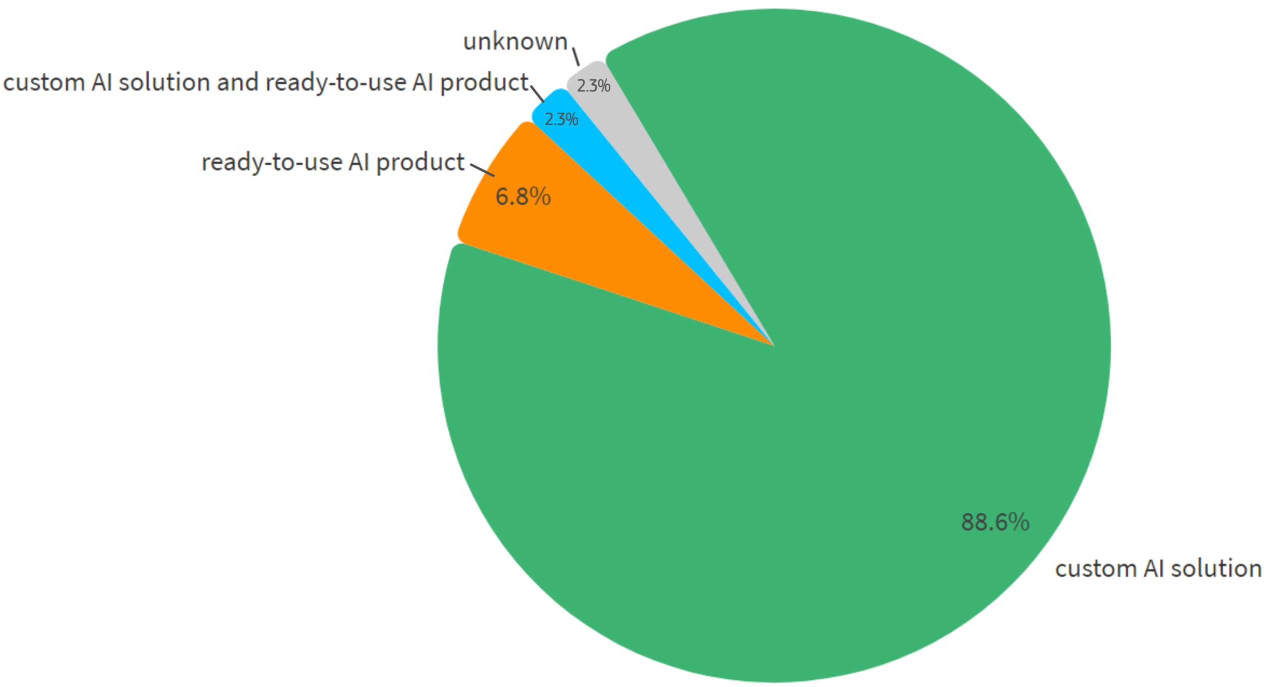
The pie chart (Figure 6) shows that most of the publication focuses on the development of novel AI solutions (either from scratch or by refinement of and / or further development of existing algorithms) (80.7%). Development covers situations where pre-trained models were subjected to “transfer learning”. Figure 6 shows that 3.4% of the studies refer to the development and use of an AI model / system. The use of existing / already developed AI model / systems represent 4.5% of the eligible studies. The use and evaluation of an AI model represent 9.1% of the eligible studies. Finally, few articles (2.3%) are about the development, use and evaluation of an AI model / system.

3.1.2 AI solution availability

We identified information regarding the AI solution availability that was developed or used in the research article. To this aim, we used the pre-defined descriptors, i.e. AI solution that include custom AI or ready-to-use AI product / commercial or a combination of the two (see method section 2.2.1).

This pie chart in Figure 7 shows that the majority of AI solution being developed in the application area 1 correspond to custom AI (88.6%). Ready-to-use products represent 6.8% of the solutions while 2.3% correspond to or a combination of custom AI and ready-to-use AI products. In some cases (2.3%), the AI solution is not specified (unknown).

Figure 7. AI solution availability



Source: JRC/Own production

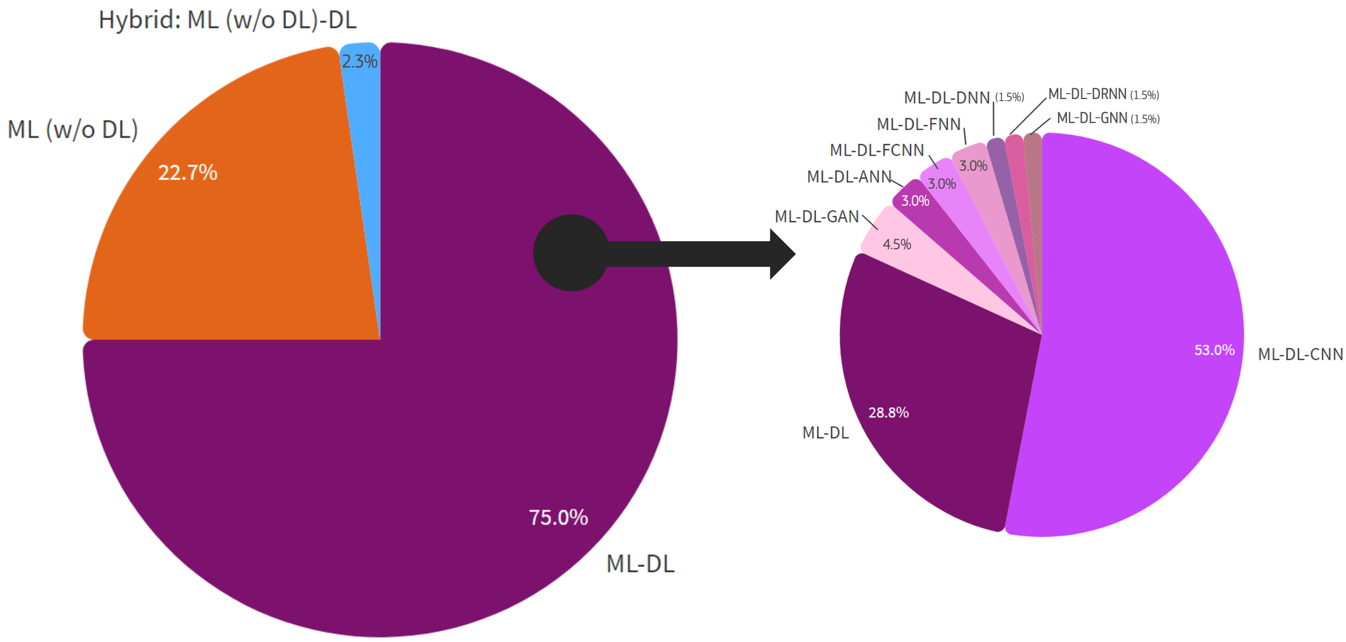
3.1.3 AI technique

We found that the majority (75%) of AI techniques being developed or used for diagnosis and prediction-based diagnosis correspond to **deep learning (ML-DL)**, (Figure 8). Interestingly, among DL models, some techniques were not specified (28.8%). Convolutional neural networks (CNN) were used most commonly for disease diagnosis i.e. algorithms that can analyse medical images and assist healthcare providers in identifying and diagnosing diseases (53.0%). In addition, other neural network models are used, such as artificial neural networks (ANN, 3.0%), deep neural networks (DNN, 1.5%), Fully connected neural network (FCNN, 3.0%), feedforward neural network (FNN, 3.0%), deep recurrent neural network (DRNN, 1.5%), graph neural network (GNN, 1.5%) and generative adversarial network (GAN, 4.5%).

Machine learning without deep-learning (ML w/o DL), i.e. what can be considered "white box" models (Burkart & Huber, 2021), followed deep learning (ML-DL) with 22.7%. The most common "white box" machine learning models used in the literature analysed were random forest classifier, logistic regression, fuzzy logics, gradient boosting machines, decision tree, K nearest neighbour (KNN), support vector machines (SVM). These were also used in various combinations.

A small percentage (2.3%) was found to correspond to **hybrid techniques** used combinations of machine learning without deep learning (ML w/o DL) and deep-learning (DL), denoted as "ML w/o DL-DL".

Figure 8. AI technique employed in the AI model / system



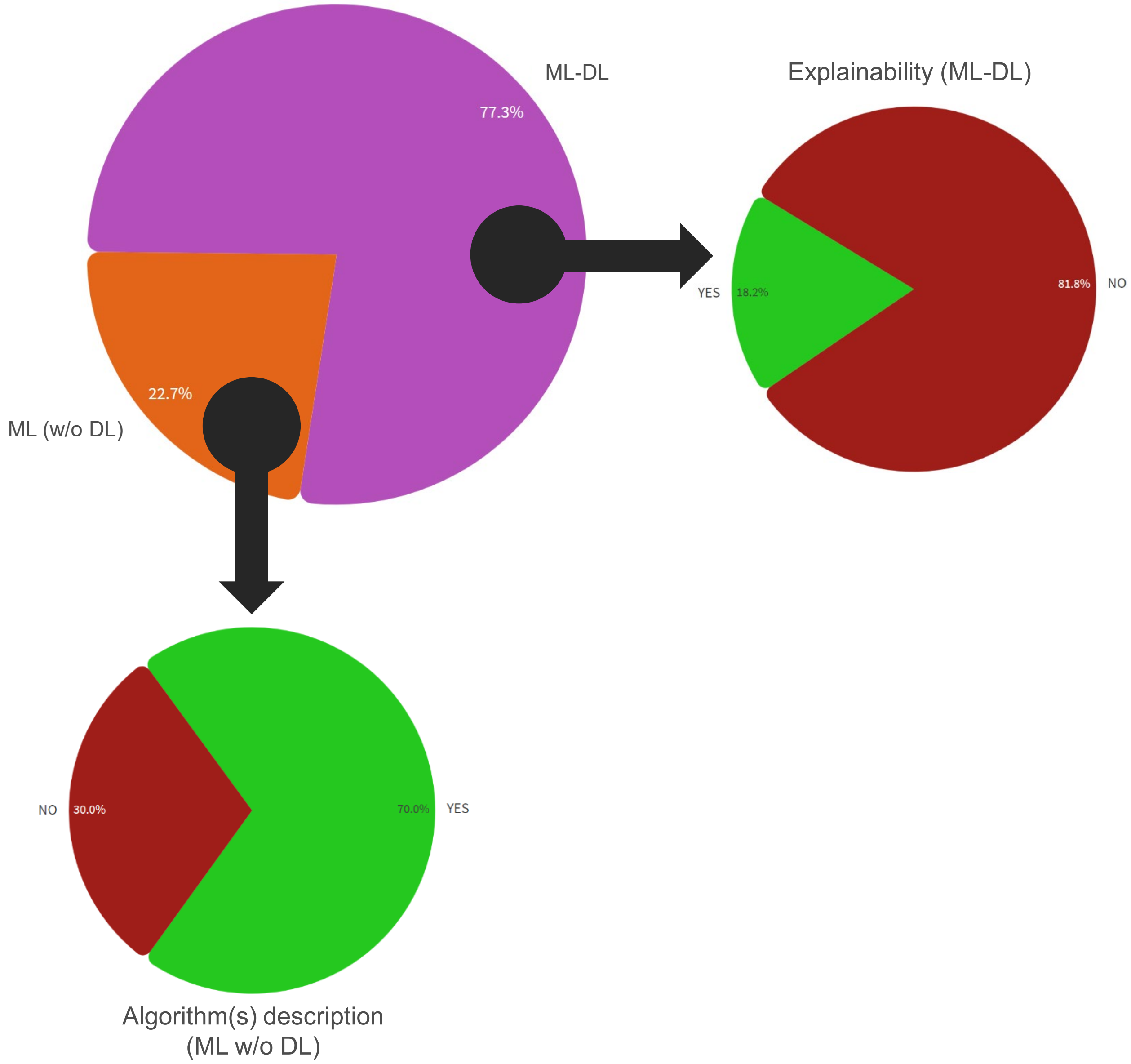
Source: JRC/Own production

3.1.4 Interpretability / explainability of the AI model / system

Next we analysed to which extent research articles addressed the concepts of interpretability or explainability of the AI system presented or used in the publications. To this end, we used the approach described in the methodology section 2.2.1.

22.7% of the studies used “white-box” machine learning (ML w/o DL) approaches. These models have the advantage of having a certain degree of intrinsic interpretability, requiring, however, clarity on what type of model was employed. Thus, we examined whether there was sufficient information on the model and algorithm(s) used in the research article. We found that 70.0% of the studies clearly describe the algorithm or a combination of algorithms employed in the AI model / system while 30.0% of the studies provided scarce or incomplete information on the algorithm(s) with repercussions on interpretability (Figure 9).

Figure 9. Interpretability / explainability of the AI model / system



Source: JRC/Own production

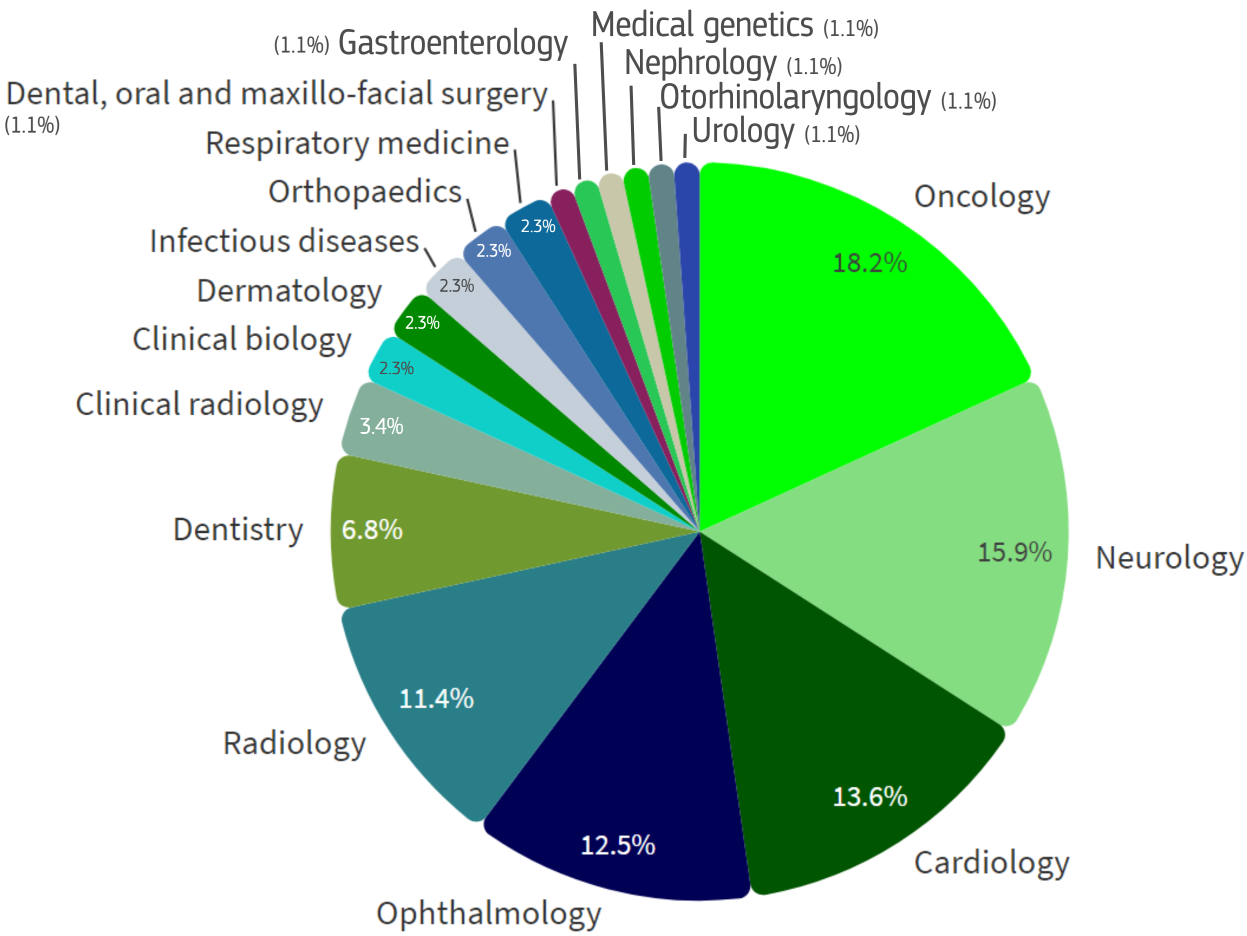
The majority of studies (77.3%) employed deep learning (ML-DL) that are intrinsically non-interpretable or 'inscrutable'. These studies also included hybrid AI models / systems.

We carefully scrutinized for this group of publications whether there was information on explainability and relevant explainability techniques. As illustrated in Figure 9, the majority of studies (81.2%) do not address explainability of the 'inscrutable' models they present. Only 18.2% of eligible studies refer to the explainabilty of the AI model / system and/or describe additional explainability tools.

3.1.5 Medical field of application

Information was identified and collected regarding the medical field for which the AI model / system was developed, used or evaluated according to the research article. To this aim, we used our customised list with pre-defined medical fields (see methodology section 2.2.1) in order to match one of them with the retrieved information from the research article.

Figure 10. Medical field of application



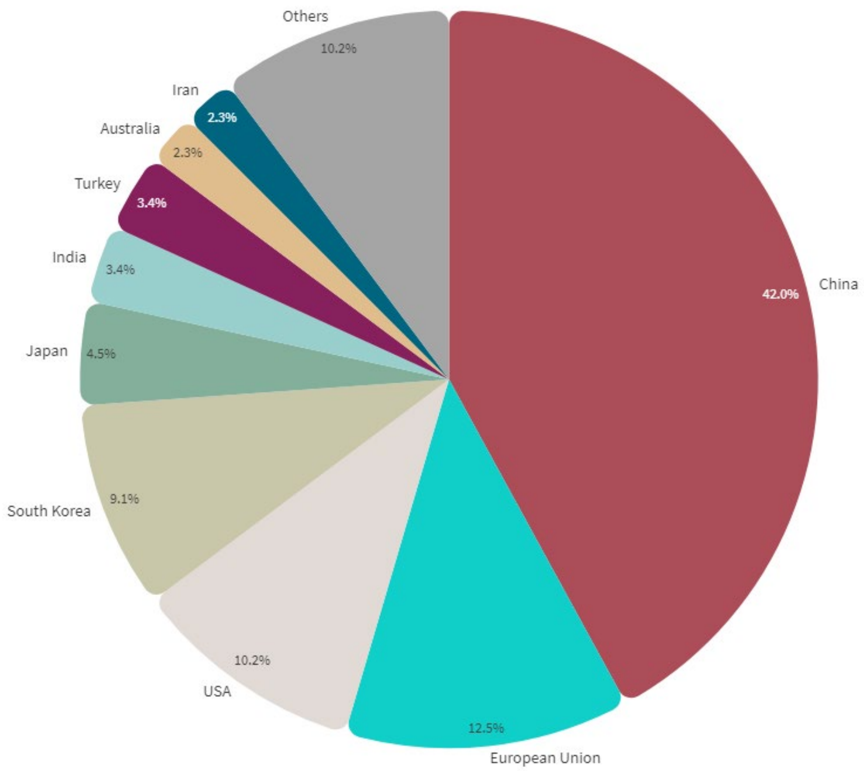
Source: JRC/Own production

In our study, oncology was found to be the most common (18.2%) medical field of application of AI models / systems with regard to application area 1. It was followed by neurology (15.9%), cardiology (13.6%), ophthalmology (12.5%), radiology (11.4%) and dentistry (6.8%). Other medical fields represent less than 4% of the studies (clinical biology, clinical radiology, dental, oral and maxillo-facial surgery, dermatology, gastroenterology, infectious diseases, medical genetics, nephrology, orthopaedics, otorhinolaryngology, respiratory medicine and urology). The total set of captured medical fields of application of AI models / systems in our study along with their representation percentages in the examined research articles are presented graphically in Figure 10.

3.1.6 Geographic origin of the publication

The distribution of scientific publications across country may indicate competitiveness and innovativeness and competitiveness in the diagnosis area. To this aim, we collected and analysed information regarding the country of the authors and / or their affiliation. With regard to the geographical distribution of the affiliated research institutions, it was found that the eligible research articles originated from 9 countries (89.8%). Major contributors were found to be China (42.0%), the EU (12.5%), the USA (10.5%), South Korea (9.5%), Japan (4.5%), India (3.4%), Turkey (3.4%), Australia (2.3%) and Iran (2.3%). Other contributors represent 10.2% (Figure 11).

Figure 11. Geographic origin of the publication



Source: JRC/Own production

3.2 Pillar 2: Analysis of reporting of key parameters

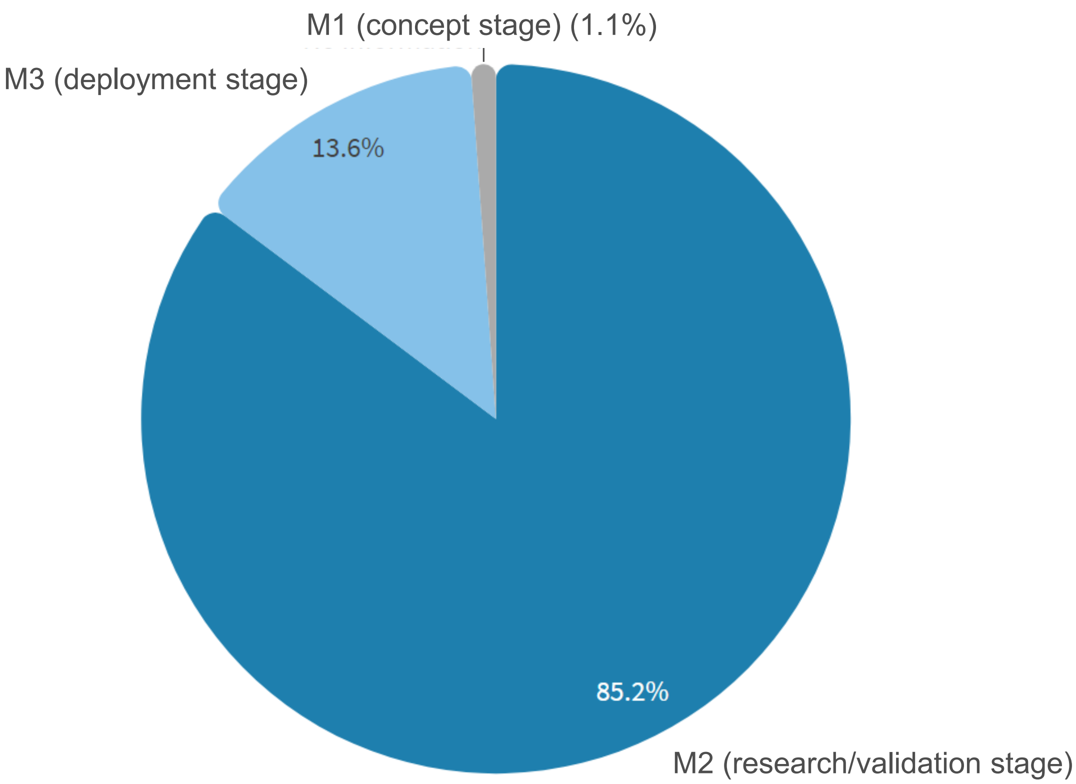
As already mentioned, the analysis of the reporting of the key parameters based on the interrogation questions is not intended to be part of this report.

However, we feel the need to present here the outcome of the interrogation question A.6 regarding the reporting of information on the technical maturity of the AI technique presented in each research article as we would like to connect these findings with the related outcome of our subjective assessment on the technical maturity level of the same AI technique.

Furthermore, in the case studies that will follow in the next section 3.3 (see Pillar 3) of this report someone can see a part of the data collected via the fiche template (**Annex 5**) regarding the reporting of the key AI parameters based on the 18 interrogation questions.

Our statistical analysis showed that 96.6% of the eligible research articles provided clear information on the technical maturity of the presented AI technique, while only 2.3% provided information with uncertainties. Interestingly, the technical maturity of the AI model / system is not explicitly mentioned in 1.1% of the cases (Figure 12).

Figure 12. Technical maturity of the AI model / system



Source: JRC/Own production

Furthermore, we also focused on the outcome of our subjective assessment regarding the classification of the technical maturity level of the presented AI technique / solution in the research articles based on the methodological approach as previously explained in the methods section 2.2.2.

According to the information analysed from each research article, we tentatively classified the corresponding technical maturity of the AI model / system in three stages, i.e. concept stage (M1), research and validation stage (M2) or deployment stage (M3).

Based on this classification model our study showed that the vast majority (85.2%) of the described AI model / systems correspond to research / validation stage (M2), 3.6% to the deployment stage (M3), while 1.1% correspond to the concept stage (M1) (Figure 12).

3.3 Pillar 3: Analysis of case studies

We made a selection of case studies from which relevant information was extracted / analysed. The "fiche" template used for collecting and documenting the relevant information extracted from the selected research articles can be found in **Annex 5**.

3.3.1 Case study 1: Development of an AI model / system

The majority of studies include the training, validation and testing of an AI model / system. Development also covers situations where pre-trained models were subjected to "transfer learning". Transfer learning is a strategic approach within ML wherein a model developed for a particular task is adapted for a second task (Yu X et al (2022)).

For example, we selected the study of Shihabuddin AR et al (2023) which is about the development of an AI model for the detection of mitotic cells in breast cancer histopathology images. It discusses the use of transfer learning as a technique to address the lack of labelled data for training machine learning models in the field of medical imaging. The study describes how pre-trained convolutional neural networks (CNNs) such as VGG16, ResNet50, and DenseNet201 are employed as part of the transfer learning process. These pre-trained models are used to extract features from histopathological images, serving as a method to transfer knowledge from the source domain (where the pre-trained models were originally trained) to the target domain (mitosis detection in medical images). This approach allows the model to leverage the learned representations from the source domain to improve its performance on the target task, despite having limited labelled data.

Bibliographic information

Title of the publication / study	**Multi CNN based automatic detection of mitotic nuclei in breast histo-pathological images**
Authors	Abdul Rahim Shihabuddin, Sabeena Beevi K.
doi	10.1016/j.compbiomed.2023.106815

Study information

Rationale / summary	In breast cancer diagnosis, the number of mitotic cells in a specific area is an important measure. It indicates how far the tumor has spread, which has consequences for forecasting the aggressiveness of cancer. Since mitosis is a complicated biological process, current methods cannot reliably identify them. The whole process is time consuming, and observations fluctuate based on the examiner's expertise, which affects the final grading process. Alternative methods like the DL approach are

	adopted. Convolutional neural network (CNN) works better in analysing images and learns good features from these images automatically. The rationale of the research lies in addressing the challenges of accurately detecting mitotic cells. The study recognises the limitations of existing methods, such as inaccuracies in identifying mitotic cells due to shape alterations and undetectable characteristics. To overcome these limitations, it adopts a DL approach, leveraging pre-trained CNNs and transfer learning to improve the accuracy and precision of mitosis detection. The proposed multiCNN framework, combined with feature reduction techniques and classification using different classifiers, aims to enhance the robustness and generalisation capability of mitosis detection methods. The application of stain normalisation, feature extraction using pre-trained CNNs, and fine-tuning of networks contribute to the comprehensive approach employed in the study. In summary, the research investigates the performance of a multi-CNN framework in mitosis detection using datasets such as MITOS-ATYPIA-14 and TUPAC16. The application of PCA is shown to enhance performance metrics, and the research highlights the potential of the proposed method in accurately detecting mitotic cells in breast cancer histopathology images.
Findings and conclusion	The multi-CNN framework, when combined with linear SVM, yields superior precision and F1-score compared to other combinations, highlighting the efficacy of the proposed method in mitosis detection. The application of PCA contributes to the improvement in performance metrics, as evidenced by the reduction in the depth of feature vectors and the capture of 95% data variance. The research demonstrates varied performance metrics when applying the proposed method to different datasets, emphasizing the impact of stain variations and structural differences on the detection process. The study provides a comprehensive comparison of the proposed method with existing works in the literature, highlighting the challenges and complexities of mitosis detection and the need for dataset-specific methodologies. The conclusions drawn from the research emphasize the potential and significance of the proposed multi-CNN framework in addressing the complexities of mitosis detection in breast cancer diagnosis. Based on the information provided, it appears that the proposed multi-CNN framework demonstrates promising results in the detection of mitotic cells in breast cancer histopathology images. The use of pre-trained convolutional neural networks (CNNs), transfer learning, and feature reduction techniques demonstrates a sophisticated and comprehensive approach to mitosis detection in histopathological images. The study also outlines potential future research directions, such as extending the classification problem to different phases of mitosis and exploring the use of additional pre-trained CNNs within the multi-CNN framework for further feature extraction.
Limitations	- The system's performance may vary based on the specific characteristics and challenges present in different datasets, as observed in the comparison of results between MITOS-ATYPIA-14 and TUPAC16 datasets. - Difficulty in making fair comparisons with other methods in the literature due to differences in datasets and methodologies may limit the broader assessment of the system's performance.

	- While the system leverages advanced DL techniques, it may still require domain expertise for interpretation and validation of results, especially in the context of medical imaging and cancer diagnosis. - It is important to consider the limitations and the need for further evaluation and validation of the system in real-world clinical settings.

Appraisal

Aboutness	Tick the appropriate box(es) ☒ **Development** ☐ **Use** ☐ **Evaluation**
Technical maturity level of the AI model /system	Tick the box ☐ **M1 = Concept** ☒ **M2 = Research / development** ☐ **M3 = Deployment**
AI technique	Convolutional neural network (CNN): AI model based on DL, specifically utilising CNNs. It uses a pre-trained CNN models (VGG16, ResNet50, DenseNet201) for feature extraction and transfer learning. Additionally, the application of PCA (Principal Component Analysis) is used for feature reduction, and the classification of mitotic cells is accomplished using classifiers such as LinearSVC, Random Forest, and Adaboost
AI solution availability	Combination of custom AI and ready-to-use product: use of ready-to-use AI products in the form of pre-trained CNNs such as VGG16, ResNet50, and DenseNet201 (pre-trained CNNs on large-scale image datasets such as ImageNet). Additionally, the use of PCA for feature reduction and classifiers like LinearSVC, Random Forest, and Adaboost can be considered as components of custom AI technique
Interpretable model OR non-interpretable but explainable AI (XAI)	No
Medical field(s)	*Primary medical field* **Clinical biology** *Secondary medical field* **Oncology**
Key parameters	A) AI model / system foundation Entity: **Clear information** Conceptual foundation: **Clear information** Intended use: **Clear information** Use scenario: **Information with uncertainties** Validation: **Clear information** Technical maturity: **Information with uncertainties**

	B) Medical fields Tick the box ☐ No information ☐ Information with uncertainties ☒ Clear information C) Users, operators, monitoring tools and metrics Potential users: Clear information User competencies: No information Training requirements: No information Performance monitoring tools: No information D) Data Modality of processed data: Clear information Data provenance: Clear information Data privacy: Clear information Curation of training data: Clear information Sources of bias: No information E) AI technique, AI model and output AI technique: Clear information Output of Ai model / system: Clear information

3.3.2 Case study 2: Development and use of an AI model / system

Regarding the development and use of an AI model / system, we can mention the study of Park C et al (2023) which is about the development and application of an AI model / system for predicting **tumor-infiltrating lymphocyte (**TIL) enrichment in the tumour microenvironment (TME) of non-small cell lung cancer (NSCLC) patients. The study uses radiomic features extracted from computed tomography (CT) images and an AI-powered TIL analysis in hematoxylin and eosin (H&E) stained images to objectively assess TIL enrichment. The main focus of the publication is on the development and application of a single AI model / system to predict TIL enrichment and its association with the outcomes of immune checkpoint inhibitor (ICI) treatment in NSCLC patients.

Bibliographic information

Title of the publication / study	**Tumor-infiltrating lymphocyte enrichment predicted by CT radiomics analysis is associated with clinical outcomes of non-small cell lung cancer patients receiving immune checkpoint inhibitors**
Authors	Park C, Jeong DY, Choi Y, Oh YJ, Kim J, Ryu J, Paeng K, Lee SH, Ock CY, Lee HY

doi	10.3389/fimmu.2022.1038089

Study information

Rationale / summary	The aim of the study is to develop a radiomic model that predicts tumor-infiltrating lymphocytes (TIL) enrichment in non-small cell lung cancer (NSCLC) using computed tomography (CT) images. The study aims to overcome the limitations of tissue-based assessment of TIL analysis and to investigate the association of predicted TIL with the outcomes of immune checkpoint inhibitors (ICI) in NSCLC patients. The study uses an AI-powered TIL analysis in hematoxylin and eosin (H&E) images and analyses its association with quantitative radiomic features extracted from CT images. The development of the radiomic model aims to provide a more precise and spatial understanding of TIL enrichment and its potential as a biomarker for ICI response in NSCLC. The radiomic model used two radiomic features, gray level variance (GLV) and large area low gray level emphasis (LALGLE), both computed from the size-zone matrix, to predict TILes. These features reflect intralesional texture heterogeneity and are associated with TIL enrichment in the tumor microenvironment. In summary, the study demonstrates the potential of radiomics in predicting TIL enrichment and its association with ICI outcomes in NSCLC. The development of an AI model for TIL analysis could assist in clinical decisions regarding ICI treatment and overcome the limitations of tissue-based analysis.
Findings and conclusion	The model is found to significantly correlate with the outcomes of ICI in NSCLC patients. Patients with high predicted TILes have significantly prolonged progression-free survival compared to those with low predicted TILes. The AI model uses two radiomic features to predict TILes. These features reflect intralesional texture heterogeneity and are associated with TIL enrichment in the tumor microenvironment. Predicted TILes is significantly associated with progression-free survival independent of PD-L1 status, a well-known biomarker for ICI response. In conclusion, the study addresses an important issue in cancer treatment by developing a radiomic model for predicting TIL enrichment and its association with ICI outcomes in NSCLC patients. It leverages AI-powered TIL analysis in H&E images and quantitative radiomic features extracted from CT images, providing a novel approach to understanding the tumor microenvironment. The study suggests the need for further validation and exploration of the temporal and spatial heterogeneity of tumors using radiomic features. The findings have potential clinical implications, as the radiomic model could assist in decision-making regarding ICI treatment and overcome limitations of tissue-based analysis.
Limitations	- The study is retrospective, which may introduce inherent biases and limitations in data collection and analysis. - The study is conducted with a limited number of patients and images, which may affect the generalizability of the findings. - The validation cohort also has a limited number of patients, which could reduce the robustness of the conclusions and limit the ability to generalise the findings to larger populations. -The study did not extensively validate the radiomic model's predictions with tissue samples and tissue biomarkers, especially in advanced cancer patients. - The study was conducted at a single center, which may limit the generalisability of the results to other healthcare settings or patient populations.

Appraisal

Aboutness	Tick the appropriate box(es) ☒ **Development** ☒ **Use** ☐ **Evaluation (including post-processing)**
Technical maturity level of the AI model / system	Tick the corresponding box ☐ **M1 = Concept** ☒ **M2 = Research / development** ☐ **M3 = Deployment**
AI technique	Deep learning: use of Lunit SCOPE IO is specifically designed for use in pathology and histopathology, particularly in the analysis of tissue samples from patients. It utilizes deep learning algorithms to assist pathologists and clinicians in interpreting and diagnosing various types of tissue specimens, including those stained with hematoxylin and eosin (H&E)
AI solution availability	Ready-to-use AI product: Lunit SCOPE IO is an AI powered diagnostic solution developed by Lunit, a medical AI company. Lunit SCOPE IO is specifically designed for use in pathology and histopathology, particularly in the analysis of tissue samples from patients
Interpretable model OR non-interpretable but explainable AI (XAI)	No
Medical field(s)	*Primary medical field* **Oncology** *Secondary medical field* Choose an item.
Key parameters	A) AI model / system foundation Entity: **Clear information** Conceptual foundation: **Clear information** Intended use: **Clear information** Use scenario: **Clear information** Validation: **Clear information** Technical maturity: **Clear information** B) Medical fields Tick the box ☐ **No information** ☐ **Information with uncertainties** ☒ **Clear information** C) Users, operators, monitoring tools and metrics Potential users: **Clear information** User competencies: **No information** Training requirements: **No information**

	Performance monitoring tools: No information D) Data Modality of processed data: Clear information Data provenance: Clear information Data privacy: No information Curation of training data: Clear information Sources of bias: Clear information E) AI technique, AI model and output AI technique: Clear information Output of AI model / system: Clear information

3.3.3 Case study 3: Use of an AI model / system

The use of existing / already developed AI model / systems represent 4.5% of the eligible studies. We chose to present the work of Du Y et al (2023) which is about the use of an AI system in the diagnosis of patients with coronary artery disease (CAD). The perivascular fat attenuation index (FAI) is a novel and validated coronary CTA (coronary computed tomography Angiogram) metric that reflects coronary inflammation. It indicates spatial changes in the perivascular fat composition caused by inflammatory signals emitted by inflamed coronary arteries. The aim of this study is to demonstrate the use of perivascular fat attenuation index (FAI) in identifying and tracking coronary inflammation in a patient with IgG4-related arteritis and coronary artery disease (CAD). The system used in this case study, which includes the AI–enhanced FAI evaluation system, demonstrates its ability to accurately measure and track perivascular inflammation in the patient with IgG4-related arteritis and CAD. The system accurately measured perivascular fat attenuation index (FAI) and provided valuable information on coronary inflammation and response to treatment. The system offers a non-invasive method of evaluating pericoronary inflammation, providing valuable insights without the need for invasive procedures.

Bibliographic information

Title of the publication / study	**Perivascular Fat Attenuation Index Mapping and Tracking Immunoglobulin G4-Related Coronary Arteritis**
Authors	Du Y, Zheng Y, Zhang H, Du J, Zhang X, Xu L, Zhou Y, Pan L.
doi	10.1161/CIRCIMAGING.122.014912

Study information

Rationale / summary	In patients with coronary artery disease (CAD), the perivascular fat attenuation index (FAI) is a novel and validated coronary CTA (coronary computed tomography Angiogram) metric that reflects coronary inflammation. It indicates spatial changes in the perivascular fat

	composition caused by inflammatory signals emitted by inflamed coronary arteries. The aim of this study is to demonstrate the use of FAI in identifying and tracking coronary inflammation in a patient with IgG4-related arteritis and CAD. It aims to show the utility of perivascular FAI in evaluating the response to anti-inflammatory treatments and to provide insights into the potential use of this metric in guiding treatment decisions for patients with autoimmune disease and established CAD. In summary, the case study demonstrates the potential of perivascular FAI in measuring and mapping coronary inflammation derived from autoimmune arteritis and tracking anti-inflammatory therapeutic effects. It also emphasises the ability of perivascular FAI to guide treatment decisions for patients with autoimmune disease and established CAD.
Findings and conclusion	This work investigates the effectiveness of pre-trained multiCNNs in detecting mitotic cells. Experiments on features extracted using various combinations of pre-trained networks are conducted. Combinations of pre-trained CNNs outperform single pre-trained CNNs. The patient presents with exertional fatigue and is found to have significant CAD and IgG4-related arteritis. Treatment with immune suppressors leads to a reduction in pericoronary inflammation, as evidenced by follow-up perivascular FAI scanning and Fluorine-18 fluorodeoxyglucose positron emission tomography examination. Despite the reduction in pericoronary inflammation, the patient remains symptomatic, and further evaluation revealed 3-vessel CAD with coronary aneurysms. The combination of AI–based technology and routine coronary CTA imaging provides a non-invasive and cost-effective method of evaluating pericoronary inflammation, plaque characteristics, and myocardial ischemia simultaneously. The AI system demonstrates its ability to accurately measure and track perivascular inflammation in the patient with IgG4-related arteritis and CAD. The system accurately measured perivascular FAI and provided valuable information on coronary inflammation and response to treatment. In conclusion, the study demonstrates the potential of perivascular FAI in measuring and mapping coronary inflammation derived from autoimmune arteritis and tracking anti-inflammatory therapeutic effects. The findings suggest that perivascular FAI can guide treatment decisions for patients with autoimmune disease and established CAD. The study highlights the predictive value of perivascular FAI on cardiac mortality and its ability to track anti-inflammatory effects of statin therapy in patients with CAD. The patient's case also emphasizes the need for careful consideration and monitoring of CAD in patients with autoimmune diseases undergoing anti-inflammatory treatments. The discriminant features from the pre-trained CNNs significantly reduce false positive rates and overcome the need for large labelled data.
Limitations	- The evaluation system faces technical challenges in evaluating coronary inflammation using Fluorine-18 fluorodeoxyglucose positron emission tomography due to physiological uptake of the tracer by the myocardium. - The interpretation of the results generated by the system requires expertise and careful consideration, as the findings need to be correlated with the patient's clinical presentation and other diagnostic tests.

Appraisal

Aboutness	Tick the appropriate box(es) ☐ **Development** ☒ **Use**

	☐ Evaluation (including post-processing)
Technical maturity level of the AI model / system	Tick the box ☐ M1 = Concept ☐ M2 = Research / development ☒ M3 = Deployment
AI technique	Machine Learning: Traditional ML method is employed for evaluating FAI data.
AI solution availability	Ready-to-use AI product: ML model from Shukun Technology Co. (doi: 10.3389/fcvm.2022.822308).
Interpretable model OR non-interpretable but explainable AI (XAI)	Yes
Medical field(s)	*Primary medical field* Cardiology *Secondary medical field* Coronary artery disease
Key parameters	A) AI model / system foundation Entity: Clear information Conceptual foundation: No information Intended use: Clear information Use scenario: Clear information Validation: No information Technical maturity: Clear information B) Medical fields Tick the box ☐ No information ☐ Information with uncertainties ☒ Clear information C) Users, operators, monitoring tools and metrics Potential users: No information User competencies: No information Training requirements: No information Performance monitoring tools: No information D) Data Modality of processed data: No information Data provenance: N.A. Data privacy: No information Curation of training data: N.A.

	Sources of bias — No information E) AI technique, AI model and output AI technique — Clear information Output of Ai model / system — No information

3.3.4 Case studies 4 and 5: Use and evaluation of an AI model / system

Regarding the use and evaluation (including post-processing) of an AI model (9.1%) we can mention the following two studies. The work of Potretzke TA et al (2023) is about the implementation and assessment of an internally developed AI algorithm for magnetic resonance (MR)-derived total kidney volume (TKV) in autosomal dominant polycystic kidney disease (ADPKD) in a real-life clinical setting. It evaluates the performance of the AI algorithm when used in clinical practice, comparing it to manual editing and other methods for determining TKV. The publication mentions that the AI algorithm for TKV calculation in ADPKD was used by nephrologists who ordered MR imaging for their patients. The algorithm was integrated into the clinical workflow and reviewed by Medical Image Analysts and radiologists. It also mentions the involvement of an integrated IT team for technical deployment and education of stakeholders, including MRI-ordering clinicians, radiologists, and MR technologists.

Bibliographic information

Title of the publication / study	**Clinical Implementation of an Artificial Intelligence Algorithm for Magnetic Resonance-Derived Measurement of Total Kidney Volume**
Authors	Potretzke TA, Korfiatis P, Blezek DJ, Edwards ME, Klug JR, Cook CJ, Gregory AV, Harris PC, Chebib FT, Hogan MC, Torres VE, Bolan CW, Sandrasegaran K, Kawashima A, Collins JD, Takahashi N, Hartman RP, Williamson EE, King BF, Callstrom MR, Erickson BJ, Kline TL
doi	10.1016/j.mayocp.2022.12.019

Study information

Rationale / summary	The study is designed to address the lack of data on the real-world performance of AI algorithms in clinical settings, particularly in the field of radiology. The implementation of AI algorithms into clinical practice involves various challenges, and there is a need to understand the factors associated with their successful integration and performance. By evaluating an AI algorithm for total kidney volume (TKV) measurement in autosomal dominant polycystic kidney disease (ADPKD), the study aims to provide insights into the impact of AI algorithm implementation on patient care and to identify the key factors influencing its real-world performance. The study investigates an internally developed and previously validated AI algorithm for magnetic resonance (MR) derived TKV in ADPKD when implemented in clinical practice. The study demonstrates the successful technical deployment of the AI algorithm into the clinical workflow, emphasizing the coordination of the interdisciplinary team, education of stakeholders, and the extraction of real-life performance metrics. The AI-based TKV measurements are found to be non-inferior to previous non-AI–assisted segmentation approaches, and the algorithm's implementation

	significantly reduces the time required for image processing. The study highlights the successful technical deployment of the AI algorithm into the clinical workflow, emphasising the coordination of the interdisciplinary team, education of stakeholders, and the extraction of real-life performance metrics.
Findings and conclusion	The successful clinical implementation of the AI algorithm for TKV measurement in ADPKD demonstrates its potential impact on patient care and the reduction of image processing time. Performance is assessed by comparing AI-based and manually edited segmentations via measures of similarity and dissimilarity to ensure expected performance. Agreement for disease class between AI-based and manually edited segmentation is high. It finds that scanner information and patient demographic factors do not significantly differ between cases that pass and those that require rework. Patient demographic factors, including age, body mass index (BMI), race / ethnicity, and study imaging date, also do not significantly differ across the pass and rework pathways. However, females are overrepresented in the corrected AI pathway, with significantly lower BMI and smaller TKVs compared to males. The algorithm's performance is found to be non-inferior to previous non-AI–assisted segmentation approaches, and it significantly reduces the time required for image processing, indicating that it effectively fulfilled its intended purpose. The study provides valuable insights into the real-world performance of the AI algorithm in a large radiology clinical practice, demonstrating its effectiveness and accuracy in clinical settings. The study concludes that the performance of the AI algorithm in a large radiology clinical practice can be preserved if careful attention is paid to the validation of the algorithm during development and if the implementation environment closely matches the development conditions. The findings provide valuable insights into the successful clinical implementation of AI algorithms, demonstrating their potential impact on patient care and the reduction of image processing time. The study highlights the importance of careful validation and integration of AI algorithms into clinical practice for their successful implementation and real-world performance. It also emphasizes the importance of coordination and education among interdisciplinary stakeholders, including clinicians, technologists, and radiologists, to ensure the successful implementation and use of the AI algorithm.
Limitations	- The overrepresentation of females in the corrected AI pathway raises questions about potential biases or factors influencing the use of the rework pathway. - While the study finds no significant changes in kidney function measurements, the impact of the AI algorithm on ADPKD severity classification is limited, with only a small percentage of rework cases resulting in a change in classification. - The study's findings may be specific to the particular clinical setting and patient population studied, and the generalisability of the results to other settings or populations may be limited.

Appraisal

Aboutness	Tick the appropriate box(es) ☐ **Development** ☒ **Use** ☒ **Evaluation (including post-processing)**

Technical maturity level of the AI model / system	Tick the box ☐ **M1 = Concept** ☐ **M2 = Research / development** ☒ **M3 = Deployment**
AI technique	Convolutional neural network (CNN): the model uses DL techniques, such as CNNs for medical image analysis. CNNs are used for image segmentation tasks in medical imaging.
AI solution availability	Custom AI: the model is designed to address the specific needs and challenges associated with ADPKD diagnosis. It uses an internally developed and previously validated algorithm for magnetic resonance (MR)-derived TKV in ADPKD.
Interpretable model OR non-interpretable but explainable AI (XAI)	No
Medical field(s)	*Primary medical field* **Clinical radiology** *Secondary medical field* **Medical imaging**
Key parameters	A) AI model / system foundation Entity: **Clear information** Conceptual foundation: **Clear information** Intended use: **Clear information** Use scenario: **Clear information** Validation: **Clear information** Technical maturity: **Clear information** B) Medical fields Tick the box ☐ **No information** ☐ **Information with uncertainties** ☒ **Clear information** C) Users, operators, monitoring tools and metrics Potential users: **Clear information** User competencies: **Clear information** Training requirements: **Clear information** Performance monitoring tools: **No information** D) Data Modality of processed data: **Clear information** Data provenance: **N.A.** Data privacy: **No information** Curation of training data: **N.A.**

	Data bias	No information
	E) AI technique, AI model and output	
	AI technique	Clear information
	Output of Ai model / system	Clear information

A second example is the work of Krishna H et al (2023) which is about the use and evaluation of a deployed AI too for the evaluation of aortic stenosis (AS) severity. It focuses on the application of the Us2.ai algorithm, an AI-driven solution that automates the interpretation of echocardiograms, to accurately quantify AS severity with no human input beyond image acquisition. The study compares the measurements derived from the AI system to those obtained by experienced human readers, demonstrating strong correlation and agreement between the AI-derived AS parameters and manual measurements. It emphasizes the potential benefits of AI technology in minimizing variability, improving interpretation, and allowing for precise and reproducible identification and management of patients with AS. The AI system, Us2.ai, is designed to analyse two-dimensional and Doppler echocardiographic images obtained from patients with normal aortic valves and varying degrees of aortic stenosis (AS).

Bibliographic information

Title of the publication / study	**Fully Automated Artificial Intelligence Assessment of Aortic Stenosis by Echocardiography, Journal of the American Society of Echocardiography**
Authors	Hema Krishna, Kevin Desai, Brody Slostad, Siddharth Bhayani, Joshua H. Arnold, Wouter Ouwerkerk, Yoran Hummel, Carolyn S.P. Lam, Justin Eze-kowitz, Matthew Frost, Zhubo Jiang, Cyril Equilbec, Aamir Twing, Patricia A. Pellikka, Leon Frazin, Mayank Kansa
doi	10.1016/j.echo.2023.03.008

Study information

Rationale / summary	The aim of the study is to evaluate the potential of AI, particularly artificial neural networks (ANNs), in accurately quantifying aortic stenosis (AS) severity. By comparing AI-derived measurements to those of experienced human readers, the study aims to determine the capacity of AI to closely mimic human assessment of key hemodynamic parameters relevant to AS severity. The goal is to address limitations in current AS assessment methods, such as time-consuming and expert-dependent transthoracic echocardiography (TTE), and to provide a more efficient and accurate approach to AS evaluation. In summary, the study demonstrates that AI technology, specifically ANNs, has the capability to closely replicate human measurements of essential parameters in the adjudication of AS severity. The application of this AI technology may offer several potential benefits, including minimising interscan variability, improving interpretation and diagnosis of AS, and enabling precise and reproducible identification and management of patients with AS.
Findings and conclusion	The study finds that ANNs, specifically the Us2.ai algorithm, closely matched expert human measurements of key parameters relevant to AS severity. The Us2.ai system appears to perform as intended in automating

	the interpretation of echocardiograms, including identifying the correct views, determining cardiac structures, and generating decision support outcomes related to cardiac structure and function. The technology demonstrates strong positive correlation and agreement with manual measurements of AS parameters, including aortic valve peak velocity (Vmax), mean pressure gradient (MPG), aortic valve area (AVA), and stroke volume index (SVi) across normal aortic valves and all degrees of AS severity. The study also highlights the potential of AI to quantify AS severity without human input beyond image acquisition. Strengths of the AI system include its ability to automate the entire workflow of echocardiographic analysis and interpretation, providing accurate and reliable measurements of AS parameters. The system has the potential to minimise interscan variability, improve interpretation and diagnosis of AS, and enable precise and reproducible identification and management of patients with AS. The study concludes that ANNs have the capacity to closely mimic human measurement of relevant parameters in the adjudication of AS severity. The findings suggest that integrating AI into the assessment of AS severity has the potential to enhance clinical practice and optimise patient care in the context of valvular heart disease.
Limitations	- The exclusion of some studies due to inadequate image quality for certain measurements. In particular, the measurement of left ventricular outflow tract diameter (LVOTd) was identified as a technically challenging aspect of AS assessment, presenting a limitation for both the AI system and human readers. - The system's current inability to exclude post-extrasystolic beats may lead to some disagreement between human and AI measurements in cases with substantial beat-to-beat variability.

Appraisal

Aboutness	Tick the appropriate box(es) ☐ **Development** ☒ **Use** ☒ **Evaluation (including post-processing)**
Technical maturity level of the AI model / system	Tick the box ☐ **M1 = Concept** ☐ **M2 = Research / development** ☒ **M3 = Deployment**
AI technique	Artificial neural network (ANN): the ANN system is designed to process echocardiographic images and perform a series of quality checks before yielding automated measurements of AS parameters. The network's architecture and algorithms were trained to accurately measure key hemodynamic parameters for AS severity assessment, with the ability to closely mimic human measurements of relevant parameters in the adjudication of AS severity.
AI solution availability	Ready-to-use AI product: Us2.ai algorithms have been cleared by the US FDA following validation at the Harvard / Brigham and Women's Hospital Echo Core Lab. The platform is described as having been externally validated in diverse real-world cohorts and approved by regulatory authorities, indicating that it is a commercially available and validated AI solution for echocardiographic analysis.
Interpretable model OR non-interpretable but explainable AI (XAI)	No
Medical field(s)	*Primary medical field*

	Cardiology *Secondary medical field* Choose an item.
Key parameters	A) AI model / system foundation Entity: Clear information Conceptual foundation: Clear information Intended use: Clear information Use scenario: Clear information Validation: Clear information Technical maturity: Clear information B) Medical fields Tick the box ☐ No information ☐ Information with uncertainties ☒ Clear information C) Users, operators, monitoring tools and metrics Potential users: Clear information User competencies: Clear information Training requirements: Clear information Performance monitoring tools: No information D) Data Modality of processed data: Clear information Data provenance: N.A. Data privacy: No information Curation of training data: N.A. Data bias: No information E) AI technique, AI model and output AI technique: Clear information Output of Ai model / system: Clear information

3.3.5 Case study 6: Development, use and evaluation of an AI model / system

Finally, few articles are about the development, use and evaluation of an AI model / system. For example, the study of Shah AA et al (2023) describes an ensemble deep learning model (EDLM) for the early identification of cholangiocarcinoma. It discusses the development of the EDLM, which consists of three deep learning algorithms — Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), and Bi-Directional LSTM (BLSTM) — and its application for the detection of mutations in cholangiocarcinoma. The study also includes a comparative assessment of the performance of the EDLM and standalone deep learning techniques. It evaluates the model using various tests and statistical techniques to assess its accuracy, sensitivity, specificity, and overall performance. Therefore, the paper encompasses both the development, use and evaluation of the AI model, as well as a comparative assessment of multiple models.

Bibliographic information

Title of the publication / study	**EDLM: Ensemble Deep Learning Model to Detect Mutation for the Early Detection of Cholangiocarcinoma**
Authors	Shah AA, Alturise F, Alkhalifah T, Faisal A, Khan YD
doi	10.3390/genes14051104

Study information

Rationale / summary	The study focuses on the development of an ensemble deep learning model (EDLM) using three deep learning (DL) algorithms for the early identification of cholangiocarcinoma, a type of cancer affecting the bile duct cells. The researchers collected a comprehensive dataset containing normal and mutated gene sequences related to cholangiocarcinoma. They used advanced feature extraction techniques to prepare the dataset for training and testing. The proposed EDLM was rigorously tested using various validation techniques and evaluated using statistical measures such as accuracy, sensitivity, specificity, and correlation coefficient. The EDLM represents a significant advancement in computationally efficient methods for early cholangiocarcinoma detection, potentially streamlining the diagnostic process. This study has the potential to significantly impact the early detection and diagnosis of cholangiocarcinoma and other forms of cancer.
Findings and conclusion	The main findings of the study include the successful development and validation of an EDLM for the early identification of cholangiocarcinoma. The EDLM, comprising LSTM, GRU, and BLSTM algorithms, demonstrated high accuracy and outperformed individual deep learning models when tested using various validation techniques, indicating its effectiveness in early cancer detection. The study suggests the potential generalisability of the proposed model to other types of cancer datasets, indicating its broad applicability in cancer detection and diagnosis. In conclusion, the study asserts that the proposed EDLM represents a significant advancement in computationally efficient methods for early cholangiocarcinoma detection. The findings suggest that the EDLM could be a valuable tool in improving the accuracy and precision of cancer diagnosis. Additionally, the study anticipates the potential for future research to expand the application of the EDLM to identify and diagnose other type of cancer, thereby contributing to advancements in healthcare and medical diagnostics.

Limitations	- The study primarily focuses on cholangiocarcinoma, and the generalisability of the model to other types of cancer may require further validation and testing. - The real-world application and integration of the EDLM into clinical settings would require validation in diverse patient populations and healthcare systems. - The implementation of advanced diagnostic tools, such as the EDLM, would require adherence to ethical guidelines and regulatory approval. Overall, the study demonstrates the potential of the EDLM for early cancer detection, but further research and considerations are needed for its practical implementation in clinical settings.

Appraisal

Aboutness	Tick the appropriate box(es) ☒ **Development** ☒ **Use** ☒ **Evaluation (including post-processing)**
Technical maturity level of the AI model / system	Tick the box ☐ **M1 = Concept** ☐ **M2 = Research / development** ☒ **M3 = Deployment**
AI technique	Deep learning (DL): the AI model employs an ensemble deep learning technique. Specifically, it consists of an ensemble DL model (EDLM) that integrates three different DL algorithms— LSTM, GRU, and BLSTM. The use of an ensemble approach combines the strengths of these individual algorithms to create a more robust and accurate predictive model for the early identification of cholangiocarcinoma.
AI solution availability	Custom AI: the model is designed and trained for the specific application of detecting mutations in cholangiocarcinoma, it details the development and evaluation of this solution.
Interpretable model OR non-interpretable but explainable AI (XAI)	No
Medical field(s)	*Primary medical field* **Oncology** *Secondary medical field* Choose an item.
Key parameters	A) AI model / system foundation Entity: **Clear information** Conceptual foundation: **Clear information** Intended use: **Clear information** Use scenario: **Clear information** Validation: **Clear information** Technical maturity: **Clear information** B) Medical fields Tick the box ☐ **No information**

	☐ Information with uncertainties ☒ Clear information C) Users, operators, monitoring tools and metrics Potential users: No information User competencies: No information Training requirements: No information Performance monitoring tools: No information D) Data Modality of processed data: Clear information Data provenance: Clear information Data privacy: Information with uncertainties Curation of training data: Clear information Data bias: No information E) AI technique, AI model and output AI technique: Clear information Output of AI model / system: Clear information

3.4 Limitations of the study

Our study has also some intrinsic limitations mainly because of the nature of the research topic. These can be summarised as follows:

<u>Period covered</u>

The period of six months that we covered (January 1, 2023, and June 30, 2023) for our scientific literature search in Pubmed / MEDLINE could be seen as a limitation in regard to whether our findings (e.g. concering the distribution of medical fields or geographic origin) are representative. However, it needs to be considered that a systematic interrogation of the literature constitutes a considerable effort (data retrieval, duplication checking, inclusion criteria etc.) and extending the period would not have allowed us to conduct this study within the envisaged timeframe.

Furthermore, there is, in our opinion, no reason to assume that this period does not offer a representative snapshot of the R&D pipeline as evidenced by the substantial amount of articles covering a variety of medical fields, AI techniques and availability of AI solutions. It is thus not necessarily plausible to assume that a longer period, e.g. one year, would have led to significantly different results. By making our search strings publicly available, other research can make use of these for future interrogations of the relevant literature. Furthermore, consideration should be given to automate, e.g. using AI, the retrieval and filtering steps as well as a preliminary gathering of key attributes.

<u>Selection and assessment of the publication attributes and key parameters</u>

Another potential limitation relates to both the selected publication 'attributes' and 'interrogation questions'. While these have been matched to pertinent ethical issues in relation to transparency and patient safety (see table in executive summary), there remains a certain arbitrariness concerning both their selection, their phrasing and, most importantly, the analysis. Although conducted using the 'four eye principle', variations of interpretation cannot be ruled out.

As touched on above, the selected publication attributes and key parameters aimed to cover essential AI-related aspects such as data, technological readiness, and intelligibility issues, which are crucial for AI application in healthcare and medicine. These aspects address important considerations such as privacy, patient consent, understanding of AI systems by healthcare professionals, and maintaining trustful patient-HCP relationships. The subjective assessment of the interrogation questions must be considered expert judgment, carrying inevitably a degree of subjectiveness.

4. Discussion

To our knowledge, this is the first study that adopts a systematic approach in investigating the R&D pipeline for AI health applications in the area of diagnosis and prediction-based diagnosis. The prototype nature of this scientific work relies largely on the selection and systematic analysis of certain key attributes related to the global innovation landscape.

In comparison, a certain number of conducted literature reviews (including systematic reviews) use a series of questions to explore the scientific literature on the application of AI in disease diagnosis. However, most of these studies are anecdotal, providing an overview of the current state of AI in disease diagnosis and highlighting the potential benefits (e.g. improving patient outcomes, reducing human errors or enhancing clinical decision-making) as well as certain risks and challenges of using AI. For example, the systematic literature review of Kumar Y et al (2023) provides a comprehensive overview of the current state of AI in disease diagnosis and highlights the potential benefits and challenges of using AI in healthcare. It uses a series of investigations to interrogate the articles and extracted from the included publications such as study characteristics and disease diagnosis outcome measures.

In contrast, our work is a systematic approach of exploring various key attributes of AI research publications, including the positioning of the study along the R&D pipeline ("aboutness"), the AI solution availability, the type of AI technologies employed, the extent to which interpretability / explainability issues were addressed, the applicable medical field(s) and the geographic origin of the publication. Based on our analysis we show some trends and we highlight aspects related to the way of reporting on essential information requirements on AI in the area of diagnosis and prediction-based diagnosis.

The ultimate objective of this approach is to emphasize the importance of addressing the challenges and limitations of AI health applications early on in the research and development stage, rather than trying to resolve them at a later stage. In particular, our work demonstrates the importance of the proper and systematic reporting of certain key attributes of scientific publications on the applications of AI models / systems under research that are anticipated to be implemented later in the health domain (deployment stage).

This report is mainly focusing on the results of our analysis with regard to the key attributes (pillar 1) and the presentation of some interesting case studies (pillar 3).

<u>Discussion points on the outcome of the analysis of the publication key attributes</u>

1) Aboutness: stage within the R&D pipeline

Our study shows that most of the publications are about **development** of their own model or system or existing algorithms with further development, i.e. training them with specific data or using / combining / fine tuning the algorithms. Development especially covers situations where pre-trained models are subjected to transfer learning. A representative example of transfer learning is the study of Kao ZK et al (2023) is about the development of four pre-trained convolutional neural network (CNNs) models (InceptionResNetV2, InceptionV3, DenseNet169, and VGG16) for the detection and classification of temporomandibular joint disc displacement (TMJDD) from structural magnetic resonance imaging (MRI) images (field of dentistry). The pre-trained models are fine-tuned using the cropped regions of interest. This approach leverages the pre-learned features from a database (the ImageNet dataset) and adapts them to the specific task of TMJDD classification, demonstrating the use of transfer learning to improve the performance of deep learning techniques for medical image analysis.

A high percentage of publications are also on **use and evaluation** of AI model / systems. A representative example of use and evaluation is the study of Potretzke TA et al (2023) which is about the assessment and implementation of an AI algorithm for magnetic resonance (MR)-derived total kidney volume (TKV) in a real-life clinical setting (field of clinical radiology). It evaluates the performance of the AI algorithm when used in clinical practice, comparing it to other methods for determining TKV. The algorithm was integrated into the clinical workflow and reviewed by medical image analysts and radiologists. It also mentions the involvement of an integrated IT team for technical deployment and education of stakeholders, including MRI-ordering clinicians, radiologists, and MR technologists.

2) AI solution availability

Regarding the **availability of AI solutions** our work shows that the majority of AI solution being developed or used in the diagnosis area corresponds to **custom AI solutions**. Among custom models, a majority corresponds to untrained or pre-trained models (existing algorithm(s) with further development) while few cases refer to *de novo* models.

Pre-trained models can be powerful tools for building AI systems, but they can also have several issues that need to be considered such as biases in the training data. This is because pre-trained models are only as good as the data they were trained on. If the training data are biased, the model will likely learn and replicate those biases. Another issue is the lack of transparency given that pre-trained models can be complex and difficult to understand, making it challenging to interpret their decisions and identify potential biases (Volkinger K et al (2021). A representative example of pre-trained models from the case studies we analysed is the study of Jung W et al (2023) that deals with the development and use of AI models for classifying panoramic in dentistry. The study uses ResNet-152 and EfficientNet-B7 in the form of pre-trained convolutional neural network (CNN) models. These two models are part of the wide range of pre-trained models available in the field of computer vision.

However, some studies refer to the ***de novo* solutions**, in order to respond to specific needs in the area of diagnosis and prediction-based diagnosis. This observation is of particular relevance as it seems that recent advances in computing power, data harvesting, and algorithm techniques led to an important uptake on the development of novel and powerful AI models / systems in the diagnosis area. A representative example from the case studies we analysed is the study of Tang Y et al (2023) which is about the development of a novel semi-supervised medical image segmentation method in the field of radiology. It introduces a new approach that incorporates of deep learning, semi-supervised learning, adversarial training, and collaborative consistency learning within the model to improve the segmentation accuracy and reliability of supervised information in medical image analysis.

3) AI technique

AI technique description is an essential aspect of ensuring that AI models / systems are reliable and trustworthy. There are issues that might arise with AI techniques in general, and specifically in the area of diagnosis, where it is essential to understand the reasoning behind a diagnosis. Some of these issues include the lack of transparency on the AI technique developed or used. AI-based products and systems may be characterised by various degrees of opacity, making it difficult to trace the AI decision-making process. This can make it challenging to identify biases or errors in the model, and to understand why the AI model is making certain predictions.

Our study shows that the majority of AI techniques being developed or used for diagnosis correspond to **deep learning (ML-DL)**. Among DL models, a significant part does not specify the precise AI technique. Interestingly, **Convolutional neural network (CNN)** was the most commonly AI technique used in diagnosis including medical imaging. A representative example from the case studies we analysed is the study of Jung W et al (2023) which is about the development and use of CNN models for classifying panoramic images in dentistry. It involves the development of an AI model using pre-trained ResNet-152 as transfer learning model and further training and evaluation of these models using panoramic radiography data.

With regard to the **choice of the technique** it is very likely that that some studies might have prioritised more complex AI model over simpler, more established alternatives, which might have led to similar results. In the study of Jung W et al (2023), the choice of ResNet-152 correspond the deepest variant, with the maximum number of layers. More complex tasks may require deeper variants with more layers (ResNet-50/101/152: Deeper variants with more layers, while simpler tasks can be handled by shallower variants (ResNet-18/34: lightweight variants with relatively fewer layers). Investigating whether simpler models can achieve comparable results to more complex ones and exploring model simplification techniques to reduce complexity might be considered.

4) Interpretability / explainability of the AI model / system

In our study, we observed in general and inadequate attention to **interpretability / explainability** aspects, with researchers often opting for complex methods without fully considering the potential difficulties that may arise.

Regarding **traditional machine learning (ML w/o DL**) models that are **intrinsically interpretable models,** our work shows that the majority of publications clearly described the algorithm or combination of algorithm(s) used – a prerequisite for interpretability "by nature". A typical example is the study of Hoffmann J et al (2023) which is about the development and use of a machine learning (ML) model. The study underscores the need for the AI system to be transparent, providing insight into its decisions and ensuring that its predictions can be easily understood and validated by human experts. The model used (Cinderella) aims to make the decision-making process of the AI model transparent and interpretable, allowing for plausibility checking by healthcare professionals. Having access to interpretability requires that the algorithm(s) is/are properly described.

However, in the case of **deep learning techniques (ML-DL)** that are **intrinsically non-interpretable or 'inscrutable' models**, a main finding is the lack of attention to **explainability** of the AI model / system. Only few publications refer to it and describe additional explainable tools. A typical example is the study of Jung W et al (2023) which is about the development and use of convolutional neural network (CNN) models. The study discusses the use of grad-CAM to generate visual descriptions of the decisions made by the CNN-based models. The visualisation provided by grad-CAM allows for the identification of important regions in the images that contribute to the classification of normal and osteoarthritis cases in the temporomandibular joint panoramic radiographs.

5) Medical field of application

By looking at the **medical field of application**, we gain a deeper understanding of the clinical context in which the AI system will be further used. By specifying the medical field of application, the developers can especially ensure that the AI system is designed and trained to address the specific needs and challenges of that field, such as diagnostic accuracy, patient safety, and treatment efficacy.

With regard to AI health applications in diagnosis and prediction-based diagnosis (area 1) our study showed that oncology was the most common medical field of application of AI models / systems followed by neurology and cardiology. These results are considered as expected given the intrinsic depth and variety of pathologies each of these fields comprise and the importance of the AI assistance has already been acknowledged in the abovementioned medical fields. For example AI is used in oncology for personalised treatment recommendations, outcome prediction, and in the development of precision medicine for cancer care (e.g. peri-operative care plans) (Sorin M et al (2023)). AI has been successfully applied in the diagnosis and monitoring of neurological disorders, such as Alzheimer's disease (Lu D et al (2023)), and in analyzing brain imaging data (Fan C et al (2023)). Furthermore, regarding cardiology AI algorithms help in the early detection of heart diseases (Jiang X et al (2023)), analysis of echocardiograms (Krishna H et al (2023)), and risk assessment for patients.

Our study found ophthalmology to be the fourth most common medical field of application of AI models / systems which at a first glance would not be considered as very expected based on the same abovementioned grounds. However, it seems that ophthalmology is currently considered to be one of the top 7 medical fields impacted by AI according to publicly available information on the web (Asghar T, (2024); Available online: *https://www.xevensolutions.com/blog/ai-in-healthcare-transforming-7-medical-fields/#Top_7_Medical_Fields_Impacted_by_AI*). Moreover, Topol EJ (2019) previously shared that it is ophthalmology and not radiology that is leading the AI movement. AI systems are used in ophthalmology to screen and diagnose retinal diseases, including diabetic retinopathy and age-related macular degeneration.

6) Geographic origin of the publication

By looking at the **geographic origin of the publication**, we gain a better understanding on potential geographical clusters in regard to AI research and development, investments and expertise as well as on reverberations on competitiveness and innovativeness. Our study showed that China, the EU and the USA have come to dominate the AI healthcare research area in terms of number of publications. It probably reflects the AI research systems in terms of funding and active scientists in these geographic areas.

In particular, our study showed that China is by far the leading country (42%) followed by the EU (12.5%) and the USA (10.5%) with regard to the geographic origin of published original research work on AI health applications with focus on the area of diagnosis and prediction-based diagnosis. Although these results represent a limited 6-months data snapshot between January 1, 2023, and June 30, 2023they are well aligned with the results of a 2024 JRC bibliometric research which showed that China is currently the leading country (30%) in terms of number of publications in the last 5 years (2019-2023) in the broader area of AI in healthcare followed closely by EU (28%) and USA (23%). This is actually the outcome a remarkable increase that China demonstrated in the AI healthcare publication share in the last 5 years (2019-2023) compared to the previous 5-years span (2013-2018), i.e. from 19% to 30%.

5. Conclusions

This is to our knowledge the first study that identifies and analyses in a systematic way reporting aspects related to the publication of scientific research articles on the development, use and evaluation of AI models / systems in healthcare (R&D pipeline).

Our key findings in regard to the examined key attributes are:

Stage of the publication within the R&D pipeline: It is not surprising that the majority of the research articles examined focused on the development of novel AI solutions (ca 81%) (i.e. either from scratch or by refinement of existing algorithms) while a small percentage concerned both development and use of AI models / systems (ca 3%). Some articles refer to the use (ca 4%) or the use and evaluation of existing / already developed AI model / systems (ca 9%). Few articles are about the development, use and evaluation of an AI model / system (ca 2%).

AI solution availability: The majority of AI solutions reported are, what we consider, 'custom-made' (ca 88%), i.e. these do not concern commercially or otherwise available ready-to-use AI systems. We were surprised that there were not more clinical investigations on such systems. However, of the pool of custom-made solutions, a high proportion relates to untrained or pre-trained models, that are retrievable from relevant model repositories such as GitHub. Popular models include the ResNet family of neural networks that is heavily used for image analysis. Very few AI solutions were genuinely developed *de novo*. When used, these seemed to respond to very specific clinical needs (e.g. for complex cases of image segmentation).

AI technique: The majority (ca 75%) of AI solutions correspond to variation of deep learning, i.e. artificial neural networks. This is not surprising given excellent performance of deep-learning based algorithms for visual AI as employed to a large extent in diagnosis. The majority of deep learning approaches concerned convolutional neural networks (ca 53%). Importantly, the use of neural networks has consequences for intelligibility of AI outcomes: neural networks are generally not intrinsically interpretable and require ex post efforts ('explainable AI', XAI) methods (e.g. saliency maps) to comprehend why and how models arrived at a given recommendation – as opposed to another one (counterfactual reasoning).

Interpretability and explainability: despite the fact that mostly intrinsically non-interpretable or 'inscrutable' models were used, very few articles (ca 18%) appeared to tackle explainability. This has important ramifications for trust and, hence, the likelihood of uptake of such models in routine clinical workflows.

Medical fields: It is not surprising that the most dynamic medical fields were oncology (ca 18%), neurology (ca 15%), cardiology (ca 13%), ophthalmology (ca 12%), radiology (ca 11%) and dentistry (ca 7%). These fields depend on high-quality medical imaging approaches and benefit clearly from the improved accuracy and predictive capacity of AI approaches. The fact that 'radiology' is on the fifth position is not surprising given its horizontal nature. Perhaps more surprisingly, a comparably large number of publications covered ophthalmology and dentistry, while fields such as orthopaedics or internal medicine are less represented.

Geographical origin: Not surprisingly, China as a major economy with about one billion citizens, is the country with the highest number of research articles (ca 42%). This may also reflect the strain experienced by Chinese health systems in the light of the fact that China has, within the last years, become a 'hyperaged' society. The EU is the second-largest publisher of AI research articles in the diagnosis area (ca 12%), followed very closely by the USA (ca 10%). Notably other Asian economies, South Korea (ca 9%) and Japan (ca 4%), are major contributors.

Taken together, our study provides robust evidence on the dynamics of research activities in the field of diagnosis. It also shows that scientific and clinical research communities do not appear to have sufficient sensitivity to the obstacles of AI uptake, namely ethical and related patient considerations. Observations include 1) lack of clarity regarding the data used in training models, especially when it comes to transfer learning, 2) rather opaque descriptions of AI techniques, including model origin and training processes, 3) lack of justification why (often) rather complex models were preferred over simpler ones. This concerns also choices within a given family of neural networks: more complicated multi-layered models (e.g. ResNet) were chosen over simpler variants. 4) Lack of clarity and evidence on explainability investigations. This is particularly surprising given the high number of 'inscrutable black box' models used.

This study is a scientific underpinning work for trustworthy AI in healthcare. It shows that more efforts are needed in regard to bridging the user and developer communities in order to overcome trust issues in regard to AI models that, if developed with sufficient circumspection and in line with key ethical principles, might bring substantial benefits to healthcare.

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List of terms / abbreviations and description

Terms / abbreviations	Description
AI	Artificial Intelligence
AI algorithm	An algorithm is a set of instructions to be followed in calculations or other operations. An AI algorithm is the programming that tells the computer how to learn to operate on its own.
AI model	An AI model is a program that has been trained on a set of data to recognise certain patterns or make certain decisions without further human intervention. An AI model employs an algorithm, a combination of algorithms or layers of algorithms.
AI system	An engineered system that generates outputs such as content, forecasts, recommendations or decisions for a given set of human-defined objectives (Definition EN ISO / IEC 22989:2023 – Artificial intelligence concepts and terminology). AI system means a machine-based system that is designed to operate with varying levels of autonomy and that may exhibit adaptiveness after deployment, and that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments (Definition AI Act).
AI technique	AI techniques refer to a set of methods and algorithms used to develop intelligent systems that can perform tasks requiring human-like intelligence. These techniques encompass various approaches like machine learning (ML), natural language processing (NLP), computer vision, and deep learning (DL).
Artificial neural network (ANN)	ANN is a subset of DL. A neural network (also artificial neural network or neural net, abbreviated ANN or NN) is a model inspired by the structure and function of biological neural networks in animal brains.
Convolutional neural network (CNN)	CNN is a subset of DL. A CNN is a regularised type of feed-forward neural network that learns features by itself via filter (or kernel) optimisation.
Data processing	The user provides the AI model datasets. The data has three processing phases: training, validation, and testing. Throughout the three phases, the AI model interprets the data to draw conclusions.

Terms / abbreviations	Description
Deep learning (DL)	DL is a type of ML based on artificial neural networks in which multiple layers of processing are used to extract progressively higher level features from data.
Deep Neural Network (DNN)	DNN is a subset of DL. DNN imitates the human brain with multiple layers for input variables to pass through. DNN is an ANN with multiple hidden layers of units between the input and output layers, which are composed of multiple linear and non-linear transformations.
Deep recurrent neural network (DRNN)	DRNN is a subset of DL. Deep RNN is a type of computer program that can learn to recognise patterns in data that occur in a sequence. It works by processing information in layers, building up a more complete understanding of the data with each layer.
Feedforward neural network (FNN)	FNN is a subset of DL. FNN is one of the two broad types of ANN, characterised by direction of the flow of information between its layers. Its flow is uni-directional, from the input nodes, through the hidden nodes (if any) and to the output nodes, without any cycles or loops, in contrast to recurrent neural networks, which have a bi-directional flow
Fully connected neural network (FCNN)	FCNN is a subset of DL. A fully connected layer refers to a neural network in which each input node is connected to each output node.
Generative adversarial network (GAN)	GAN is a subset of DL. GAN is a deep learning architecture. It trains two neural networks to compete against each other to generate more authentic new data from a given training dataset.
Graph neural network (GNN)	GNN is a subset of DL. GNNs are designed to perform inference on data described by graphs. GNNs are neural networks that can be directly applied to graphs, and provide an easy way to do node-level, edge-level, and graph-level prediction tasks.
Intended purpose	Intended purpose means the use for which an AI system is intended by the provider, including the specific context and conditions of use, as specified in the information supplied by the provider in the instructions for use, promotional or sales materials and statements, as well as in the technical documentation (Definition AI Act).

Terms / abbreviations	Description
Machine learning (ML)	The use and development of computer systems that are able to learn and adapt without following explicit instructions, by using algorithms and statistical models to analyse and draw inferences from patterns in data.
MEDLINE	MEDLINE is the National Library of Medicine's (NLM) premier bibliographic database that contains more than 31 million references to journal articles in life sciences with a concentration on biomedicine. MEDLINE is a primary component of PubMed and includes literature published from 1966 to present, as well as a selected coverage of literature prior to that period. The PubMed database has been available since 1996 and contains more than 37 million citations and abstracts of biomedical literature. Citations in PubMed primarily stem from the biomedicine and health fields, and related disciplines such as life sciences, behavioral sciences, chemical sciences, and bioengineering.
MeSH	Medical Subject Headings - MeSH is the NLM controlled vocabulary thesaurus used for indexing articles for PubMed.
PubMed	PubMed is a free resource supporting the search and retrieval of biomedical and life sciences literature with the aim of improving health–both globally and personally. The PubMed database contains more than 37 million citations and abstracts of biomedical literature.
Testing data	Data used for providing an independent evaluation of the AI system in order to confirm the expected performance of that system before its placing on the market or putting into service (Definition AI Act).
Training data	Data used for training an AI system through fitting its learnable parameters (Definition AI Act).
Validation data	Data used for providing an evaluation of the trained AI system and for tuning its non-learnable parameters and its learning process in order, inter alia, to prevent underfitting or overfitting (Definition AI Act).

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Annexes

Annex 1. Source of information - Pubmed / MEDLINE database

According to the US National Library of Medicine webpage (www.nlm.nih.gov / bsd / difference.html) the following scientific information content is publicly available from the Pubmed / MEDLINE database:

"MEDLINE is the National Library of Medicine's (NLM) premier bibliographic database that contains more than 31 million references to journal articles in life sciences with a concentration on biomedicine. MEDLINE is a primary component of PubMed and includes literature published from 1966 to present, as well as a selected coverage of literature prior to that period. A distinctive feature of MEDLINE is that the records are indexed with NLM Medical Subject Headings (MeSH). MeSH is the NLM controlled vocabulary thesaurus used for indexing articles for PubMed.

PubMed is a free resource supporting the search and retrieval of biomedical and life sciences literature with the aim of improving health–both globally and personally. The PubMed database has been available since 1996 and contains more than 37 million citations and abstracts of biomedical literature. Citations in PubMed primarily stem from the biomedicine and health fields, and related disciplines such as life sciences, behavioral sciences, chemical sciences, and bioengineering.

PubMed's references include the MEDLINE database plus the following types of citations:

- *In-process citations, which provide records for articles before those records are indexed with Medical Subject Headings (MeSH) or converted to out-of-scope status.*
- *Citations to articles that are out-of-scope (e.g., covering plate tectonics or astrophysics) from certain MEDLINE journals, primarily general science and general chemistry journals, for which only the life sciences articles are indexed with MeSH.*
- *"Ahead of Print" citations that precede the article's final publication in a MEDLINE indexed journal.*
- *Citations that precede the date that a journal was selected for MEDLINE indexing (when supplied electronically by the publisher).*
- *Pre-1966 citations that have not yet been updated with current MeSH and converted to MEDLINE status.*
- *Citations to some additional life sciences journals that submit full text to PMC® (PubMed Central®) and receive a qualitative review by NLM.*
- *Citations to author manuscripts of articles published by NIH-funded researchers.*
- *Citations for the majority of books available on the NCBI Bookshelf (a citation for the book and in some cases each chapter of the book).*

PubMed citations often include links to the full-text article on the publishers' websites and / or in PMC and the Bookshelf. MEDLINE is the largest subset of PubMed. PubMed search retrieval can be limited to MEDLINE citations by restricting the search to the MeSH controlled vocabulary or by using the Journal Categories filter called MEDLINE.

Annex 2. Literature search strategy protocol for application area 1

Biomedical literature database: Pubmed / MEDLINE
Period: 1 / 1 / 2023 - 30 / 6 / 2023

("artificial intelligence"[MeSH Terms] **OR** ("Knowledge bases"[MeSH Terms:NoExp] **NOT** "sentiment analysis"[MeSH Terms]) **OR** "artificial intelligence"[ALL Fields])

AND

("diagnos*"[ALL Fields] **OR** ("diagnosis"[MeSH Terms] **AND** "Diagnostic techniques and procedures"[MeSH Terms:NoExp]) **OR** ("Investigative Technique*"[MeSH Terms] **NOT** ("population surveillance"[MeSH Terms] **OR** "drug development"[MeSH Terms] **OR** "drug discovery"[MeSH Terms] **OR** "vaccine development"[MeSH Terms])) **OR** "Genomic*"[MeSH Terms] **OR** "Metabolomic*"[MeSH Terms] **OR** "omics-based"[ALL Fields] **OR** "Image Processing, Computer-Assisted"[MeSH Terms] **OR** "Immunochemistry"[MeSH Terms] **OR** ("Microbiology"[MeSH Terms] **NOT** ("environmental microbiology"[MeSH Terms])) **OR** "theragnostic*"[ALL Fields] **OR** "Diagnosis, Computer-Assisted"[MeSH Terms]

AND

("diseases category"[MeSH Terms] **NOT** ("animal diseases"[MeSH Terms] **OR** "Chemically-Induced Disorders"[MeSH Terms] **OR** "congenital abnormalities"[MeSH Terms] **OR** "Fetal Diseases"[MeSH Terms] **OR** "genetic diseases, inborn"[MeSH Terms] **OR** "Disorders of Environmental Origin"[MeSH Terms] **OR** "Transfusion Reaction"[MeSH Terms] **OR** "Occupational Diseases"[MeSH Terms] **OR** "Morphological and Microscopic Findings"[MeSH Terms] **OR** "Pathologic Processes"[MeSH Terms] **OR** "Signs and Symptoms"[MeSH Terms] **OR** "Wounds and Injuries"[MeSH Terms])
OR "disease*"[ALL Fields])

AND

("Anatomy category"[MeSH Terms] **NOT** ("Animal Structures"[MeSH Terms] **OR** "Bacterial Structures"[MeSH Terms] **OR** "Embryonic Structures"[MeSH Terms] **OR** "Fungal Structures"[MeSH Terms] **OR** "Animal scales"[MeSH Terms] **OR** "Plant Structures"[MeSH Terms] **OR** "Viral Structures"[MeSH Terms])
OR
"Anatom*"[ALL Fields])

NOT

(("healthcare"[ALL Fields] **OR** "health care"[ALL Fields] **OR** "Delivery of Health Care"[MeSH Terms] **OR** "Health Care Sector"[MeSH Terms] **OR** "Comprehensive Health Care"[MeSH Terms] **OR** "Internal Medicine"[MeSH Terms] **OR** "Clinical Medicine"[MeSH Terms] **OR** "Pediatrics"[MeSH Terms] **OR** "Surgical Procedures, Operative"[MeSH Terms] **OR** "Medical Informatics"[MeSH Terms])
OR
("health research*"[ALL Fields] **OR** "drug development"[MeSH Terms] **OR** "drugs, investigational"[MeSH Terms] **OR** "vaccine research*"[ALL Fields] OR "vaccine development"[MesH Terms] **OR** "compassionate use"[ALL Fields] **OR** "health records, personal"[MeSH Terms])
OR
("health system*"[ALL Fields] **OR** "health services"[MeSH Terms] **OR** "health services administration"[MeSH Terms] **OR** "health planning"[MeSH Terms] **OR** "health management"[ALL Fields] **OR** "healthcare management system"[ALL Fields] **OR** "health system* management"[ALL Fields] **OR** "healthcare system*"[ALL Fields] **OR** "telemedicine"[MeSH Terms] **OR** "equipment and supplies"[MeSH Terms] **OR** ("health system"[ALL Fields] **AND** "supply chain"[ALL Fields]) **OR** "Health Informatics"[ALL Fields])
OR
("public health"[ALL Fields] **OR** "disease outbreak*"[ALL Fields] **OR** "epidemiolog*"[ALL Fields] **OR** "public health"[MeSH Terms] **OR** "public health surveillance"[MeSH Terms] **OR** "public health informatics"[MeSH Terms])))

AND

(2023 / 1 / 1:2023 / 6 / 30[pdat])

Annex 3. Interrogation questions for the reporting of key parameters

	Question	Answer level		
	A. AI model / system: foundations			
A.1	**Entity** *Is there information about the entity responsible for developing and / or providing the AI model / system?*	No information	Information with uncertainties	Clear information
A.2	**Conceptual foundations / scientific relevance** *Is there information about the scientific evidence in relation to the AI model / system's design and its intended use?*	No information	Information with uncertainties	Clear information
A.3	**Intended use** *Is the intended use of the AI model / system described?*	No information	Information with uncertainties	Clear information
A.4	**Use scenario / environments** *Is there information on the specific use scenario(s) and use environment(s) in which the AI model / system is intended to operate?*	No information	Information with uncertainties	Clear information
A.5	**Validation** *Is there information on the extent of initial quality evaluation of the AI model / system?*	No information	Information with uncertainties	Clear information
A.6	**Technical maturity** *Does the publication contain information on the technical maturity of the AI model / system?*	No information	Information with uncertainties	Clear information
	B. Field of application			
B.1	**Medical fields and / or research areas** *Is there information on the medical field(s) and / or research areas to which the AI model / system would be applied / contribute?*	No information	Information with uncertainties	Clear information
	C. Users, operators, monitoring tools and metrics			
C.1	**Potential users** *Are the potential users of the AI model / system clearly identified?*	No information	Information with uncertainties	Clear information
C.2	**User competency** *Is the required user competency described?*	No information	Information with uncertainties	Clear information

C.3	**Training requirements** *Are the indications on training requirements for the competent and safe intended use of the AI model / system described?*	No information	Information with uncertainties	Clear information
C.4	**Performance monitoring tools / metrics** *Is there information on the availability of performance monitoring tools / metrics that should be used by the operators of the AI model / system?*	No information	Information with uncertainties	Clear information
	D. Data			
D.1	**Modality of processed data** *Is there information on data modality or modalities processed by the AI model / system?*	No information	Information with uncertainties	Clear information
D.2	**Data provenance of training data** *Is there information on the provenance of the data employed for training the AI model / system?*	No information	Information with uncertainties	Clear information
D.3	**Privacy / handling of personal data** *Is there information on whether and how personal data / privacy issues have been addressed during development or use of the AI model / system?*	No information	Information with uncertainties	Clear information
D.4	**Curation of training data** *Is there information on the curation of the data employed for training the AI model / system?*	No information	Information with uncertainties	Clear information
D.5	**Potential sources of data bias** *Are potential sources of data bias discussed in the publication?*	No information	Information with uncertainties	Clear information
	E. AI technique and output			
E.1	**AI technique** *Is the AI technique employed clearly described in the publication?*	No information	Information with uncertainties	Clear information
E.2	**Output of AI model / system** *Is there information on the generated output(s) of the AI model / system?*	No information	Information with uncertainties	Clear information

Annex 4. Developed list of medical fields of application

1	*Anaesthetics*
2	*General surgery*
3	*Neurosurgery*
4	*Obstetrics and gynecology*
5	*Internal medicine*
6	*Ophthalmology*
7	*Otorhinolaryngology*
8	*Paediatrics*
9	*Respiratory medicine*
10	*Urology*
11	*Orthopaedics*
12	*Pathology*
13	*Neurology*
14	*Psychiatry*
15	*Clinical radiology*
16	*Radiotherapy*
17	*Laboratory medicine*
18	*Biological hematology*
19	*Microbiology*
20	*Clinical chemistry*
21	*Clinical biology*
22	*Immunology*
23	*Plastic surgery*
24	*Thoracic surgery*
25	*Paediatric surgery*
26	*Vascular surgery*
27	*Cardiology*
28	*Gastroenterology*
29	*Rheumatology*

30	*General hematology*
31	*Endocrinology*
32	*Physical medicine and rehabilitation*
33	*Stomatology*
34	*Neuro-psychiatry*
35	*Dermato-venerology*
36	*Dermatology*
37	*Venereology*
38	*Radiology*
39	*Tropical medicine*
40	*Child psychiatry*
41	*Geriatrics*
42	*Nephrology*
43	*Infectious diseases*
44	*Public health and Preventive Medicine*
45	*Pharmacology*
46	*Occupational medicine*
47	*Allergology*
48	*Gastro-enterologic surgery*
49	*Nuclear medicine*
50	*Accident and emergency medicine*
51	*Clinical neurophysiology*
52	*Maxillo-facial surgery*
53	*Dental, oral & maxillo-facial surgery*
54	*Podiatric Surgery*
55	*Podiatric Medicine*
56	*General Practice*
57	*Toxicology*
58	*Medical oncology*

Annex 5. Fiche template for collecting and documenting case study information

Bibliographic information

Title of the publication / study	
Authors	
doi	

Study information

Rationale / summary	
Findings and conclusion	
Limitations	*(e.g. early study development, readiness level, performing as intended, obstacles, etc.)*

Appraisal

Aboutness	Tick the appropriate box(es) ☐ **Development** ☐ **Use** ☐ **Evaluation (including post-processing)**
Technical maturity level of the AI model / system	Tick the box ☐ **M1 = Concept** ☐ **M2 = Research / development** ☐ **M3 = Deployment**
AI technique	Please specify the type(s) of AI technique is employed in the AI model / system
AI solution availability	Please specify the AI solution that is being developed or used. The solution may include custom AI or ready-to-use AI product / commercial or a combination of the two
Interpretable model OR non-interpretable but explainable AI (XAI)	Does the study mention explainable AI (XAI)?
Medical fields and / or research areas	*Primary medical field* Choose an item. *Secondary medical field* Choose an item. *Other research area(s)* Please specify
Key parameters	A) AI model / system foundation

Entity	Choose an item.
Conceptual foundation	Choose an item.
Intended use	Choose an item.
Use scenario	Choose an item.
Validation	Choose an item.
Technical maturity	Choose an item.

B) Medical fields and / or research areas
Tick the box

☐ **No information**
☐ **Information with uncertainties**
☐ **Clear information**

C) Users, operators, monitoring tools and metrics

Potential users	Choose an item.
User competencies	Choose an item.
Training requirements	Choose an item.
Performance monitoring tools	Choose an item.

D) Data

Modality of processed data	Choose an item.
Data provenance	Choose an item.
Data privacy	Choose an item.
Curation of training data	Choose an item.
Sources of bias	Choose an item.

E) AI technique, AI model and output

AI technique	Choose an item.
Output of Ai model / system	Choose an item.

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